Machine Learning Elective Project: Boston House Price Prediction Low Code Report

MIT Applied AI and Data Science Program

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Executive Summary

The dataset analyzed in this project is the Boston Housing dataset, collected in 1970 from the Boston Standard Metropolitan Statistical Area (SMSA). The attributes (independent variables, features, or predictors) in the dataset are evaluated as essential predictors of home prices in Boston suburbs and towns. A multiple linear regression model is built to effectively capture variation in Boston home prices across neighborhoods and vicinities. The analysis in this project comprises exploratory data analysis, feature engineering, and thorough diagnostics for multicollinearity to build a robust, accountable ordinary least squares (OLS) multiple linear regression model.

Exploratory data analysis revealed robust spatial and socioeconomic heterogeneity. Independent variables such as the percentage of the local population in lower-status (LSTAT), the average number of rooms per dwelling (RM), and weighted distances to five Boston employment centers (DIS) revealed strong relationships with home values. Houses in neighborhoods farther from Boston employment centers were more likely to have large-lot zoning (ZN). In contrast, houses in areas with a higher index of accessibility to radial highways (RAD) usually had higher property-tax rates (TAX). These independent variables shaped housing prices according to the importance of location, zoning, and population demographics. Outlier values in each independent numerical variable were retained because they reflect real-world conditions.

To address skewness and improve the linearity of the target-dependent median value (MEDV) for owner-occupied homes, a log transformation was applied. The index of accessibility to radial highways (RAD) was one-hot encoded to avoid absolute multicollinearity. In contrast, the Charles River dummy variable (CHAS) remained a binary variable. A Variance Inflation Factor (VIF) analysis was conducted to identify and remove superfluous dummy variables with infinite or undefined VIF values, resulting in an appropriate matrix design for an ordinary least squares (OLS) multiple linear regression model. The remaining independent variables (features) with acceptable VIF values showed no multicollinearity. The variables with higher VIF values indicate accurate socioeconomic clustering rather than structural prolixity.

The final OLS multiple linear regression model corroborated the robustness of the information content and its alignment with economic intuition. Given the multiple

linear regression model's R-squared of about 0.70, it explains approximately 70% of the variation in home prices. Since the root mean squared error (RMSE), mean absolute error (MAE), and mean absolute percent error (MAPE) are small and stable, the multiple linear regression model's predictions are valid. Since the training and test dataset performance does not show evidence of overfitting, the multiple linear regression model generalizes well. The OLS multiple linear regression model reveals that the most influential predictors or key drivers (i.e., independent variables or features) that determine Boston house value property are per capita crime rate by town (CRIM), adjacency to the Charles River (i.e., CHAS), nitric oxide concentration (NOX), average number of rooms per dwelling (RM), weighted distances to five Boston employment centers (DIS), pupil-teacher ratio by town (PTRATIO), and percentage of lower status of the population (LSTAT). The CRIM predictor indicates that higher crime rates are associated with lower home values. The CHAS predictor indicates that homes located near the Charles River are valued more than homes not located near the Charles River. The NOX predictor indicates that homes located near high nitric oxide pollution have lower values. The RM predictor indicates that homes with more abundant rooms have higher values. The DIS predictor indicates that homes farther from employment centers have lower values. The PTRATIO predictor indicates that homes located near poor-performing overcrowded schools with higher pupil-teacher ratios are associated with lower home values. The LSTAT predictor indicates that homes in suburbs or towns with a high percentage of lower-status populations generally have lower values.

Altogether, this project provides a definite data-driven discernment of the independent variables, or features, that affect the pricing of Boston homes. The project also delivers a statistically robust OLS multiple linear regression model designed for effective interpretation and governance. The overall findings from this project highlight the interdependence among environmental, zoning policy, transportation access, and neighborhood demographic factors that affect the Boston residential market in 1970. Real estate business individuals, home buyers, property investors, and housebuilders can utilize the OLS multiple linear regression model to make operational decisions, pricing strategy decisions, investment dispositions, marketing emphasis decisions, and risk evaluation decisions.

Business Problem Overview and Solution Approach

Business Problem Overview

The business problem for this project is that different organizations and their relevant stakeholders (e.g., home buyers, property investors, and housebuilders) need to understand which factors determined house values in Boston suburbs, towns, and neighborhoods in 1970. Without the usage of a data-driven approach in determining how zoning, environmental conditions, accessibility, and demographic factors affect Boston housing prices, many relevant stakeholders (e.g., home buyers, property investors, and housebuilders) are at risk of incorrectly allocating resources, mispricing properties, and incorporating policies that do not capture achievable economic and community standards. Thus, the core of the business problem is to correctly categorize the key performance variables or features that deterministically determine housing prices in Boston. Consequently, the relevant stakeholders (e.g., home buyers, property investors, and housebuilders) would be able to make informed and vital decisions about the development, investment, and policy design for homes in Boston in 1970.

Analysis Explanation in Achieving Business Objective

The business objective of this project is to accurately predict housing prices in Boston across different towns and suburbs using the compiled locality features in the dataset. To meet this objective, it is critical to identify the key factors driving house prices in these distinct Boston suburbs and towns. The analysis thus supports this business objective by transforming housing and neighborhood data into legitimate insights. By applying exploratory data analysis methods, techniques of data preprocessing, and framing an OLS multiple linear regression model, the project overall assesses how each factor (e.g., per capita crime rate (CRIM), proportion of residential land zoned (ZN), proportion of non-retail business acres per town (INDUS), Charles River traction (CHAS), nitric oxide concentration parts per 10 million (NOX), number of rooms per dwelling (RM), proportion of owner-occupied units built before 1940 (AGE), weighted distances to five Boston employment centers (DIS), index of accessibility to radial highways (RAD), full-value property-tax rate per $10,000 (TAX), pupil-teacher ratio by town (PTRATIO), and percentage of lower status of the population (LSTAT)) affect the main target median value of owner-occupied homes in $1,000 (MEDV). The analysis used in this project ensures that the ordinary least squares (OLS) multiple linear regression model is interpretable and valid after

addressing issues such as skewed distributions, categorical encoding, and multicollinearity. These actionable insights help stakeholders (e.g., home buyers, property investors, and housebuilders) in these organizations understand which independent variables, or features, most significantly affect housing prices in Boston, and identify the necessary interventions or investments that will have the greatest impact.

Solution Approach or Methodology

The project follows a data-driven, multi-faceted methodology, using Python-based statistical modeling and computations conducted by a Data Science team to address the business problem and objective, including critical issues, key insights, operational findings, strategies, and recommendations from the provided dataset. The approach emphasizes Python-powered statistical analyses (e.g., code snippets are provided in the report's appendix) to deliver comprehensive data analysis and insights to support strategic interventions and recommendations. As the dataset is collected from the Boston Standard Metropolitan Statistical Area (SMSA) in 1970, different organizations and relevant stakeholders (e.g., home buyers, property investors, and housebuilders) can use a house-prediction model (e.g., a carefully specified OLS multiple linear regression model) that accurately describes home prices in Boston during that year. At the same time, the model can provide essential, valuable information to specific home buyers, property investors, and housebuilder stakeholders. In the same manner, since the dataset captures housing prices from 1970, these prices can best reflect the current market valuation of homes, which can help determine the validity of today's house prices. The model can also help potential buyers choose the housing characteristics they prefer within their budget. The solution approach also includes a combination of descriptive analytics and correlation analysis. Additional steps include: data cleaning; log-transforming the target MEDV variable to improve linearity; one-hot encoding the RAD independent variable; using VIF diagnostics to remove redundant categorical dummy variables; using a train-test split to evaluate model performance; and removing non-statistically significant independent variables. The project methodology yields a stable, high-quality model that best explains variation in housing prices while simultaneously providing an infrastructure for key decision-making in real estate planning, pricing, and policy development.

Data Overview

In this section, the structure of the data from the provided dataset is presented. The dataset drawn from the Boston Standard Metropolitan Statistical Area (SMSA) in 1970 includes 506 unique homes (each represented by a row).

Categorical Variables

The collected dataset includes two categorical independent variables. The first is a dummy binary categorical variable, known as the "Charles River dummy variable" (CHAS), which takes two possible values (1 if there is river adjacency near the Boston home and 0 if there is none). The second categorical variable, the "Index of accessibility to radial highways," abbreviated as RAD, uses numeric codes to represent distinct categories of home accessibility to radial highways. These codes include values of 1, 2, 3, 4, 5, 6, 7, 8, and 24.

Numerical Variables

The collected dataset includes 10 distinct, independent, continuous numerical variables. These include "Per capita crime rate by town," or abbreviated as CRIM, "Proportion of resdiential land zoned for lots over 25,000 square feet," or abbreviated as ZN, "Proportion of non-retail business acres per town," or abbreviated as INDUS, "Nitric Oxide concentration pollution (parts per 10 million)," or abbreviated as NOX, "Average number of rooms per dwelling," or abbreviated as RM, "Proportion of owner-occupied units built before 1940," or abbreviated as AGE, "Weighted distances to five Boston employment centers," or abbreviated as DIS, "Full-value property-tax rate per $10,000," or abbreviated as TAX, "Pupil-teacher ratio by town," or abbreviated as PTRATIO, and "Percentage of lower status of the population," or abbreviated as LSTAT. The target variable, or the dependent variable in this collected dataset, is "Median value of owner-occupied homes in $1,000," or abbreviated as MEDV.

Understanding the Structure of the Data from the Dataset (See Appendix for Code Snippets)

Table 1. Sample preview of the First 5 Rows in the dataset

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	LSTAT	MEDV
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	4.98	24.0
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	9.14	21.6
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	4.03	34.7
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	2.94	33.4
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	5.33	36.2

As noted earlier in the description of both categorical and numerical variables, the table shows each variable featured in separate columns from left to right as: CRIM (e.g., Per capita crime rate by town), ZN (e.g., Proportion of resdiential land zoned for lots over 25,000 square feet), INDUS (e.g., Proportion of non-retail business acres per town), CHAS (e.g., Charles River dummy variable), NOX (e.g., Nitric Oxide concentration pollution–parts per 10 million), RM (e.g., Average number of rooms per dwelling), AGE (e.g., Proportion of owner-occupied units built before 1940), DIS (e.g., Weighted distances to five Boston employment centers), RAD (e.g., Index of accessibility to radial highways), TAX (e.g., Full-value property-tax rate per $10,000), PTRATIO (e.g., Pupil-teacher ratio by town), LSTAT (e.g., Percentage of lower status of the population), and MEDV (e.g., Median value of owner-occupied homes in $1,000–target variable).

Table 2. Sample preview of the Last 5 Rows in the dataset

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	LSTAT	MEDV
501	0.06263	0.0	11.93	0	0.573	6.593	69.1	2.4786	1	273	21.0	9.67	22.4
502	0.04527	0.0	11.93	0	0.573	6.120	76.7	2.2875	1	273	21.0	9.08	20.6
503	0.06076	0.0	11.93	0	0.573	6.976	91.0	2.1675	1	273	21.0	5.64	23.9
504	0.10959	0.0	11.93	0	0.573	6.794	89.3	2.3889	1	273	21.0	6.48	22.0
505	0.04741	0.0	11.93	0	0.573	6.030	80.8	2.5050	1	273	21.0	7.88	11.9

Similarly, the last five rows in the dataset follow the same tabular structure as the first five rows, beginning with CRIM and ending with MEDV.

Shape of the Dataset (See Appendix for Code Snippet)

Code Output 1. Shape of the Dataset

(506, 13)

Observations on the Shape of the Dataset

The dataset has 506 rows and 13 columns. The collected dataset provides a relatively very small number of observations (e.g., rows) for each column (e.g., independent variables, including the target dependent variable).

Data Types found in the Dataset (See Appendix for Code Snippet)

Table 3. Column Variables with associated Data Types

```
Data columns (total 13 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   CRIM     506 non-null    float64
 1   ZN       506 non-null    float64
 2   INDUS    506 non-null    float64
 3   CHAS     506 non-null    int64
 4   NOX      506 non-null    float64
 5   RM       506 non-null    float64
 6   AGE      506 non-null    float64
 7   DIS      506 non-null    float64
 8   RAD      506 non-null    int64
 9   TAX      506 non-null    int64
 10  PTRATIO  506 non-null    float64
 11  LSTAT    506 non-null    float64
 12  MEDV     506 non-null    float64
dtypes: float64(10), int64(3)
```

Observations on Dataset Types

The dataset provided has 13 columns, all numeric. It is important to indicate that the CRIM, ZN, INDUS, NOX, RM, AGE, DIS, PTRATIO, LSTAT, and MEDV columns are all float types, which are generally represented as decimals. The remaining columns, CHAS, RAD, and TAX, are all integer types, generally represented as whole numbers. To reiterate, the CHAS and RAD columns are dummy and numeric code-type categorical variables that take only whole-number values. Unlike the CHAS and RAD

variables, the TAX column uses an integer type because it contains meaningful numeric rate representations.

Missing Values in the Dataset (See Appendix for Code Snippet)

Table 4. Number of Missing Values in the Dataset

	0
CRIM	0
ZN	0
INDUS	0
CHAS	0
NOX	0
RM	0
AGE	0
DIS	0
RAD	0
TAX	0
PTRATIO	0
LSTAT	0
MEDV	0

Observations on Missing Values in the Dataset

No missing values were found across all columns in the dataset. This finding suggests that the collected dataset was successfully compiled. In other words, the analysis in this report is polished, yielding reliable insights and conclusions that support accurate, justified recommendations.

Duplicate Values in the Dataset (See Appendix for Code Snippet)

Code Output 2. Number of Duplicate values in the Dataset

np.int64(0)

Observations on Duplicate Values in the Dataset

No duplicate entries were found across all columns in the dataset. This finding further suggests that the collected dataset was accurately collected. Again, this proves that the analysis in this report is reputable and provides reliable insights and conclusions, resulting in accurate and justified recommendations.

Summary Descriptive Statistics for Numerical Non-categorical Variables (See Appendix for Code Snippet)

Table 5. Summary Descriptive Statistics for all Variables

	count	mean	std	min	25%	50%	75%	max
CRIM	506.0	3.61	8.60	0.01	0.08	0.26	3.68	88.98
ZN	506.0	11.36	23.32	0.00	0.00	0.00	12.50	100.00
INDUS	506.0	11.14	6.86	0.46	5.19	9.69	18.10	27.74
CHAS	506.0	0.07	0.25	0.00	0.00	0.00	0.00	1.00
NOX	506.0	0.55	0.12	0.38	0.45	0.54	0.62	0.87
RM	506.0	6.28	0.70	3.56	5.89	6.21	6.62	8.78
AGE	506.0	68.57	28.15	2.90	45.02	77.50	94.07	100.00
DIS	506.0	3.80	2.11	1.13	2.10	3.21	5.19	12.13
RAD	506.0	9.55	8.71	1.00	4.00	5.00	24.00	24.00
TAX	506.0	408.24	168.54	187.00	279.00	330.00	666.00	711.00
PTRATIO	506.0	18.46	2.16	12.60	17.40	19.05	20.20	22.00
LSTAT	506.0	12.65	7.14	1.73	6.95	11.36	16.96	37.97
MEDV	506.0	22.53	9.20	5.00	17.02	21.20	25.00	50.00

Descriptive Statistics for Per capita crime rate by town (CRIM)

The minimum per capita crime rate by town is about 0.01. The average per capita crime rate by town is about 3.61. The maximum per capita crime rate by town is about 88.98. The average per capita crime rate by town indicates lower crime exposure overall for Boston residents.

Descriptive Statistics for Proportion of residential land zoned for lots over 25,000 square feet (ZN)

The minimum proportion of residential land zoned for lots over 25,000 square feet is 0. The average proportion of residential land zoned for lots over 25,000 square feet is about 11.36. The maximum proportion of residential land zoned for lots over 25,000 square feet is 100. The average residential land zoned for lots over 25,000 square feet indicates that most towns have some land reserved for large-lot housing, whereas the majority of the land in the Boston area is zoned for smaller lots or other uses. In other words, the dataset includes both suburban and semi-urban areas, where the typical Boston town has some upscale, low-density residential zoning.

Descriptive Statistics for Proportion of non-retail business acres per town (INDUS)

The minimum proportion of non-retail business acres per town is about 0.46. The average proportion of non-retail business acres per town is about 11.14. The maximum proportion of non-retail business acres per town is about 27.74. The average proportion of non-retail business acres per town indicates that the typical Boston town in the dataset is neither heavily industrial nor purely residential. In clear terms, the average value suggests an adequate level of industrialization in suburban areas located around Boston.

Descriptive Statistics for Charles River dummy variable (CHAS)

Because this variable is binary, the only meaningful descriptive statistic is the mean. The mean is 0.07, representing the proportion of houses near the Charles River. This characteristic reflects a distinctive, uncommon environmental amenity, since most homes in neighborhoods are not river-adjacent.

Descriptive Statistics for Nitric Oxide concentration pollution–parts per 10 million (NOX)

The minimum Nitric Oxide concentration in parts per 10 million is about 0.38. The average Nitric Oxide concentration in parts per 10 million is about 0.55. The maximum Nitric Oxide concentration in parts per 10 million is about 0.87. The average Nitric Oxide concentration in the dataset suggests moderate levels, indicating air pollution dispersed across the Boston area. To clarify, the Nitric Oxide concentration suggests moderate urbanization, with no unsafe levels of air pollution from a moderate presence of highways, traffic, or industrial activity.

Descriptive Statistics for Average number of rooms per dwelling (RM)

The minimum average number of rooms per dwelling is about 3.56. The average number of rooms per dwelling is about 6.28. The maximum average number of rooms per dwelling is about 8.78. The average number of rooms per dwelling indicates that most dwellings in the Boston area are spacious, found in many suburban, middle- and upper-middle-class neighborhoods.

Descriptive Statistics for Proportion of owner-occupied units built before 1940 (AGE)

The minimum proportion of owner-occupied units built before 1940 is about 2.90. The average proportion of owner-occupied units built before 1940 is about 68.57. The maximum proportion of owner-occupied units built before 1940 is about 100. The average proportion of owner-occupied units built before 1940 indicates that many Boston neighborhoods and towns have more older homes than newer ones, which are typically found in denser urban areas rather than newer, more suburban areas.

Descriptive Statistics for Weighted distances to five Boston employment centers (DIS)

The minimum weighted distances to five Boston employment centers are about 1.13. The average weighted distances to five Boston employment centers are about 3.80. The maximum weighted distances to five Boston employment centers are about 12.13. The average weighted distance to five Boston employment centers indicates that most Boston towns are located in middle-distance suburbs, where commuting is common, and jobs are moderately accessible to residents.

Descriptive Statistics forthe Index of accessibility to radial highways (RAD)

The minimum accessibility index to radial highways is 1. The maximum accessibility index to radial highways is 24. The average accessibility index to radial highways is 9.55. Because the index of radial highway accessibility is categorical, these descriptive statistics indicate that many neighborhoods have modest highway accessibility, with only a small number having very high accessibility, suggesting these homes are near major highway junctions.

Descriptive Statistics for Full-value property-tax rate per $10,000 (TAX)

The minimum full-value property tax rate per $10,000 is about 187. The average full-value property tax rate per $10,000 is about 408.24. The maximum full-value property tax rate per $10,000 is about 711. The average full-value property tax rate per $10,000 indicates that Boston towns have a significant property tax burden, suggesting moderate municipal expenditures and a run-of-the-mill tax environment.

Descriptive Statistics for Pupil-teacher ratio by town (PTRATIO)

The minimum pupil-teacher ratio by town is about 12.60. The average pupil-teacher ratio by town is about 18.46. The maximum pupil-teacher ratio by town is about 22. The average pupil-teacher ratio by town suggests that class sizes in most Boston school locales are larger, indicating that some schools might be overcrowded, lack ideal small-group environments, and have fewer teachers per student than the student population. At the same time, the average pupil-teacher ratio suggests that the Boston teams are classified as middle-income driven learning environments.

Descriptive Statistics for the Percentage of lower status of the population (LSTAT)

The minimum percentage of the lower-status population is about 1.73. The average percentage of the lower-status population is about 12.65. The maximum percentage of the lower-status population is about 37.97. The average lower-status percentage indicates that most of the Boston residential population is middle-class. It also suggests that a small but significant portion of residents are of the lower class, while others are of the higher class.

Descriptive Statistics for Median value of owner-occupied homes in $1,000 (MEDV)

The minimum median value of owner-occupied homes in $1,000 is about 5. The average median value of owner-occupied homes in $1,000 is about 22,530. The maximum value of owner-occupied homes in $1,000 is about 50. The median value of owner-occupied homes in $1,000 suggests that a typical home in the Boston area in the 1970s was modest. In other terms, different values of higher, moderate, and lower Boston home costs indicate lower-income neighborhoods, middle-class suburbs, and higher-valued areas, respectively.

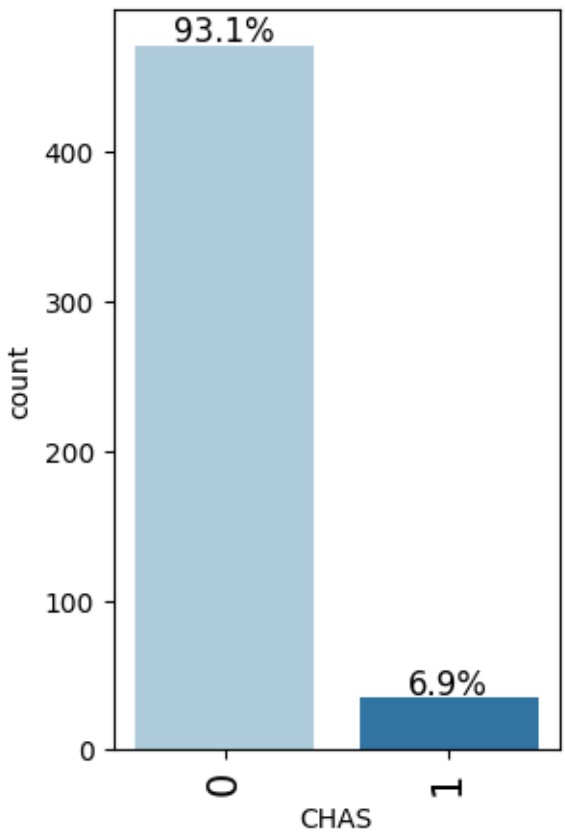
Exploratory Data Analysis (EDA) and Data Preprocessing

Univariate Analysis

A univariate analysis was conducted to examine the nature, distribution, and implications of each variable in the dataset. The median is the middle value in a dataset, splitting the data into two halves. A bar plot function was used to graph the two categorical variables: the Charles River dummy variable (CHAS) and the Index of accessibility to radial highways (RAD). A combined histogram and boxplot function was used to graph the remaining numerical variables: the Per capita crime rate by town (CRIM), the Proportion of residential land zoned for lots over 25,000 square feet (ZN), the Proportion of non-retail business acres per town (INDUS), the Nitric Oxide concentration pollution–parts per 10 million (NOX), the Average number of rooms per dwelling (RM), the Proportion of owner-occupied units built before 1940 (AGE), the Weighted distances to five Boston employment centers (DIS), the Full-value property-tax rate per $10,000 (TAX), the Pupil-teacher ratio by town (PTRATIO), the Percentage of lower status of the population (LSTAT), and the Median value of owner-occupied homes in $1,000 (MEDV). Please refer to the appendix for the code snippets for the barplot, combined histogram, and boxplot functions.

Charles River dummy variable (CHAS) (See Appendix for function call Code Snippet)

Graph 1. Charles River dummy variable Barplot distribution (CHAS)

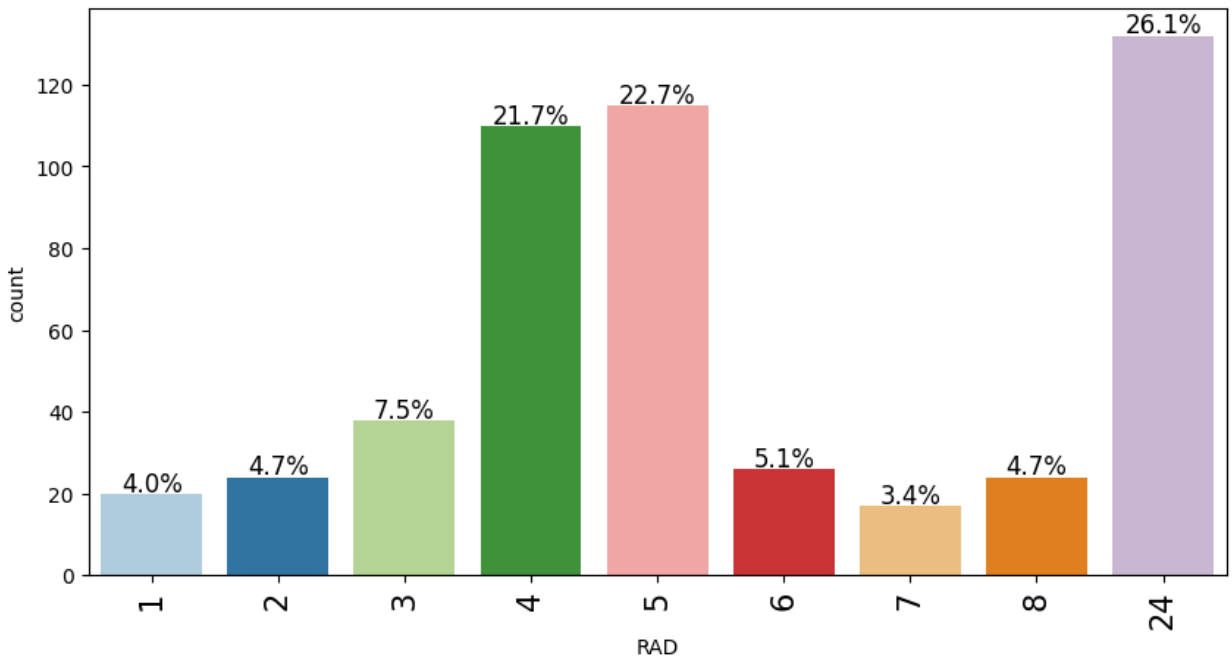


Observations and Implications on Charles River dummy variable graph

The bar plot above shows that the distribution of Boston homes adjacent to the Charles River is skewed. About 93.1% of all Boston homes are located not near the river. In contrast, about 6.9% of all Boston homes are located near the Charles River. The graph interpretation aligns with the geographical notion that a narrow strip of land lies along the Charles River. The lower percentage of homes located near the Charles River suggests that river-adjacent Boston homes are usually more desirable and may correlate with higher median home values (MEDV).

Index of accessibility to radial highways (RAD) (See Appendix for function call Code Snippet)

Graph 2. Index of accessibility to radial highways variable Barplot distribution (RAD)

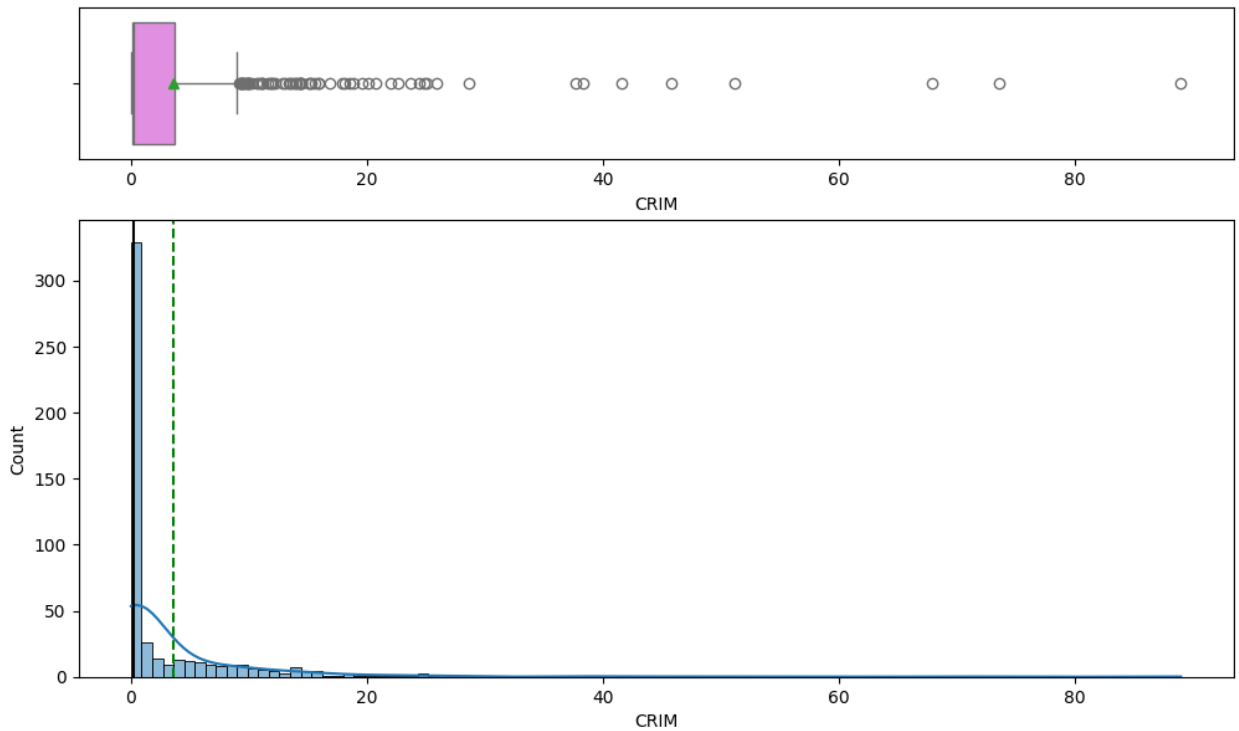


Observations and Implications on the Index of accessibility to radial highways graph

The bar plot above shows that the distribution of accessibility index values to radial highways is uneven. About 26.1% of all Boston homes are located near tracts with very notable highway accessibility, as measured by index 24. About 22.7% of all Boston homes are located near highways with an index of 5. About 21.7% of all Boston homes are located near highways with an index of 4. Altogether, these index values of 24, 5, and 4 comprise of 70% of all highway accessibility for Boston Homes. Instead, the remaining index values of 3, 6, 2, 8, 1, and 7 suggest unusually low accessibility for Boston homes near radial highways. The implications of these varying indices of accessibility to radial highways suggest that many Boston suburbs are connected to major roads. As a result, these roads could see more traffic, more commercial activity, and possibly quicker commutes with well-connected routes. Also, these higher RAD index values could suggest lower median home values due to noise and traffic, as well as higher levels of nitric oxide pollution and greater commercial and industrial zoning.

Per capita crime rate by town (CRIM) (See Appendix for function call Code Snippet)

Graph 3. Histogram and Boxplot: Per capita crime rate by town distribution (CRIM)

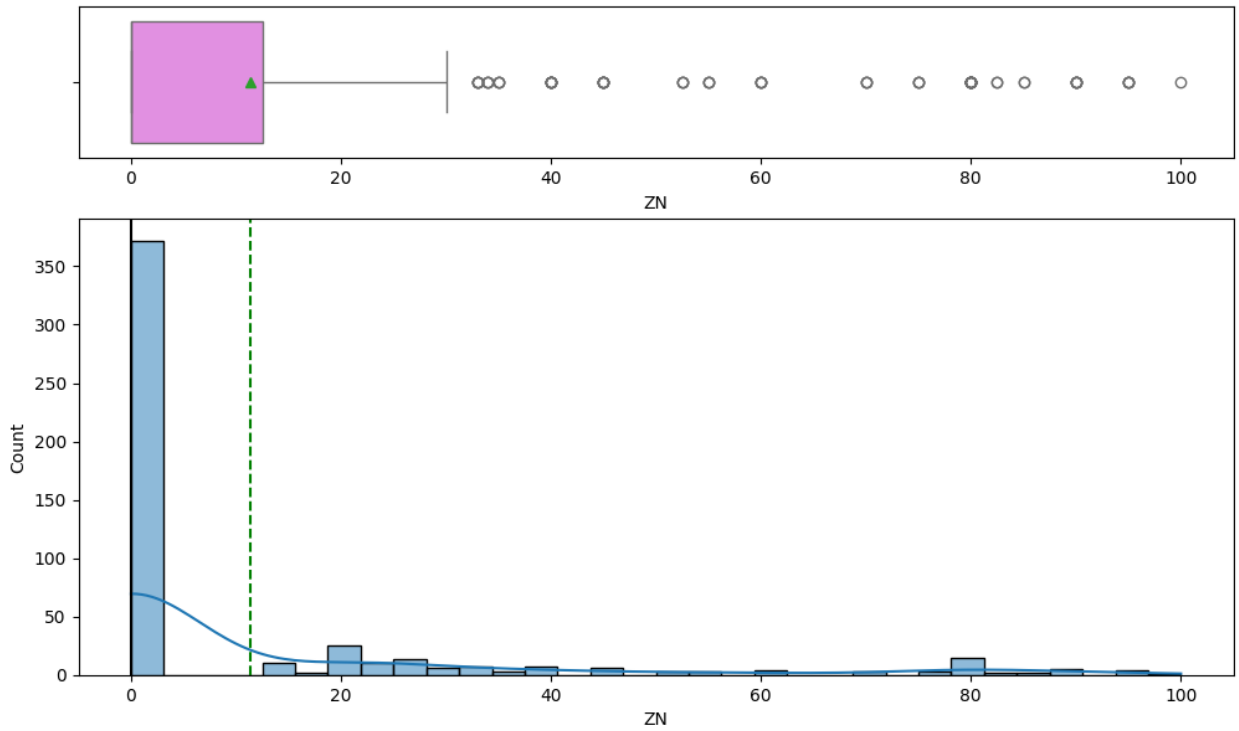


Observations and Implications on the Per capita crime rate by town histogram and boxplot

The histogram and boxplot of the per capita crime rate by town are heavily right-skewed. The majority of towns have significantly low crime rates (usually 0), whereas a small number of towns have extremely high crime rates (ranging from approximately 20 or higher). This suggests that crime rates are not evenly distributed across Boston towns, with the median (denoted by a solid black line in the histogram) at 0.26 and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) at 3.61. Most towns in the dataset have substantially low crime exposure. There are also many outliers, suggesting a few towns with substantially high crime rates. These values range from about 15 to 80. These are likely to be highly dense, urban, or socioeconomically stressed areas. The overall higher per capita crime rate is thus associated with lower median home values, a higher percentage of lower-status individuals in the population, higher nitric oxide pollution, and denser urban areas.

Proportion of residential land zoned for lots over 25,000 square feet (ZN) (See Appendix for function call Code Snippet)

Graph 4. Histogram and Boxplot: Proportion of residential land zoned for lots over 25,000 square feet distribution (ZN)



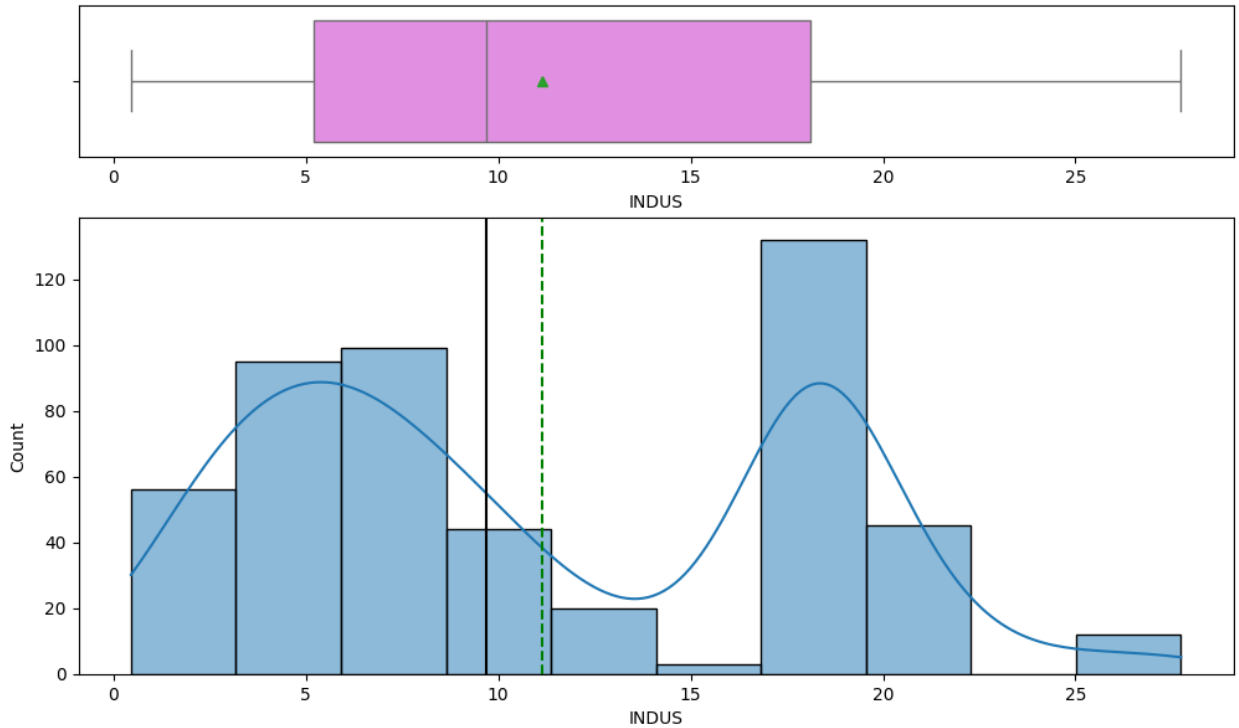
Observations and Implications on the Proportion of residential land zoned for lots over 25,000 square feet histogram and boxplot

The histogram and boxplot of the proportion of residential land zoned for lots over 25,000 square feet are heavily right-skewed. The majority of residential land zoned for lots over 25,000 square feet is 0. There are very few towns with very high ZN values, usually ranging from approximately 20 to close to 100. The median (denoted by a solid black line in the histogram) is 0, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 11.36. Both the histogram and boxplots indicate multiple outliers, suggesting significantly large-lot zoning in these towns. These outlier values range from 30 to 100. This also implies that there are more suburban, lower-density, prosperous towns where Boston homes are built on large pieces of land that are divided or designated for ownership or development. Also, this further implies that most towns are more urbanized or moderately dense, suggesting older, compacted housing patterns. Conversely, this implies that a few

towns have substantially large-lot zoning, which classifies them as more suburban and has higher median home costs.

Proportion of non-retail business acres per town (INDUS) (See Appendix for function call Code Snippet)

Graph 5. Histogram and Boxplot: Proportion of non-retail business acres per town (INDUS)



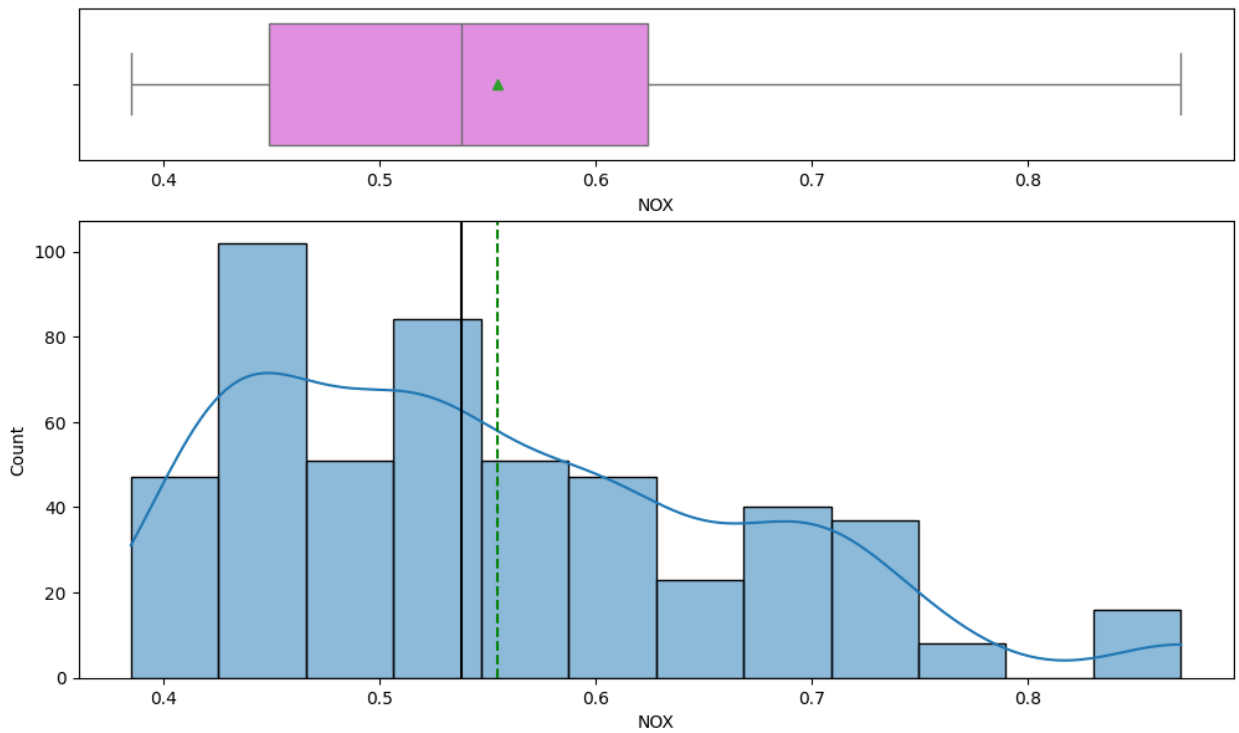
Observations and Implications on the Proportion of non-retail business acres per town histogram and boxplot

The histogram and boxplot of the proportion of non-retail business acres per town are moderately right-skewed. A significant number of towns have lower INDUS values ranging from 2 to 8. A smaller number of towns have higher INDUS values, ranging from 15 to 28. The histogram shows a concentration of towns with INDUS values in the lower-to-mid range. The median (denoted by a solid black line in the histogram) is 9.69, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 11.14. Towns with higher INDUS valuesindicatey areas with densely clustered highways, factories, or business areas. The median home value in

these areas is likely to be lower. Also, the towns with lower INDUS values indicate quieter suburban zones with more residential areas. The median home value in these areas is likely to be higher.

Nitric Oxide concentration pollution–parts per 10 million (NOX) (See Appendix for function call Code Snippet)

Graph 6. Histogram and Boxplot: Nitric Oxide concentration pollution–parts per 10 million (NOX)



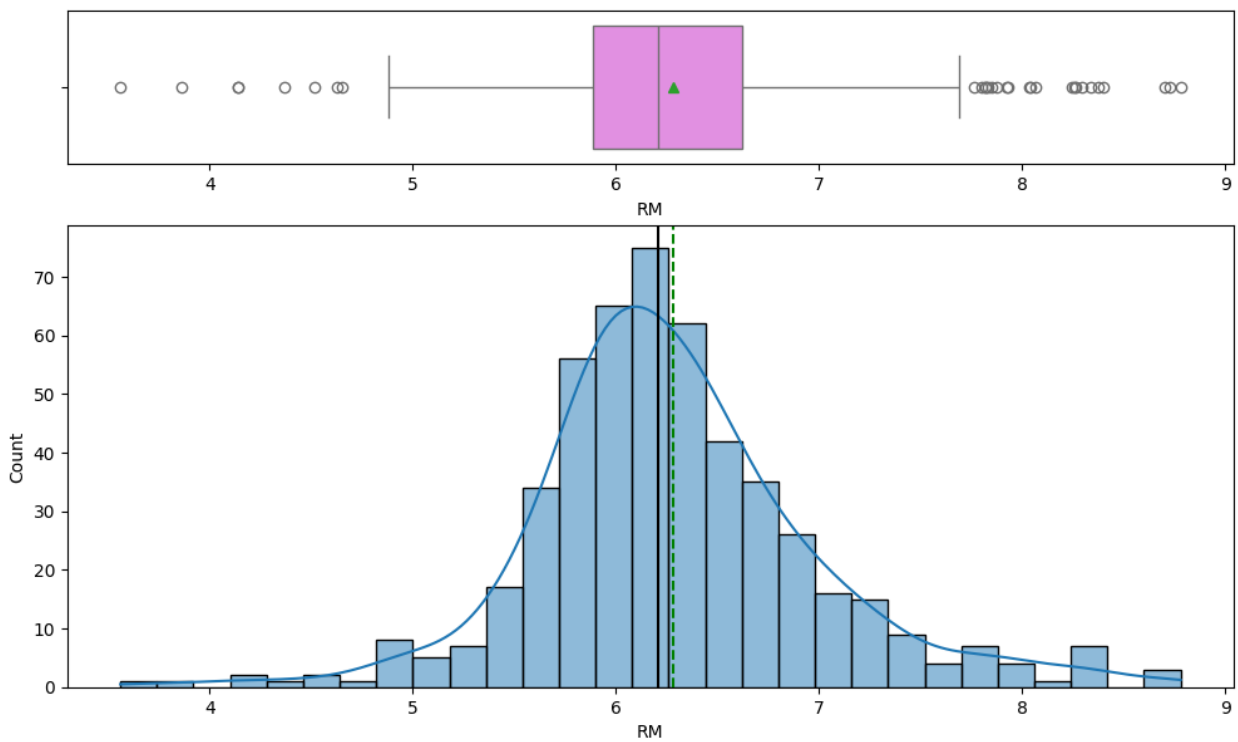
Observations and Implications on the Nitric Oxide concentration pollution–parts per 10 million histogram and boxplot

The histogram and boxplot of the nitric oxide concentration pollution are moderately right-skewed. Most Boston towns have mid-range nitric oxide pollution levels ranging from about 0.45 to 0.6. Fewer towns have higher nitric oxide pollution levels, ranging above 0.7. The median (denoted by a solid black line in the histogram) is 0.54, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 0.55. The dataset overall contains a mix of cleaner suburban areas and more polluted urban or industrial areas. Generally, Boston towns are clustered

together with similar levels of nitric oxide pollution. In relation to median home values, towns with lower nitric oxide levels are mostly associated with higher median home values because they are located in suburban areas. In contrast, towns with higher nitric oxide levels are mostly associated with lower median home values because they are located in industrial areas.

Average number of rooms per dwelling (RM) (See Appendix for function call Code Snippet)

Graph 7. Histogram and Boxplot: Average number of rooms per dwelling (RM)



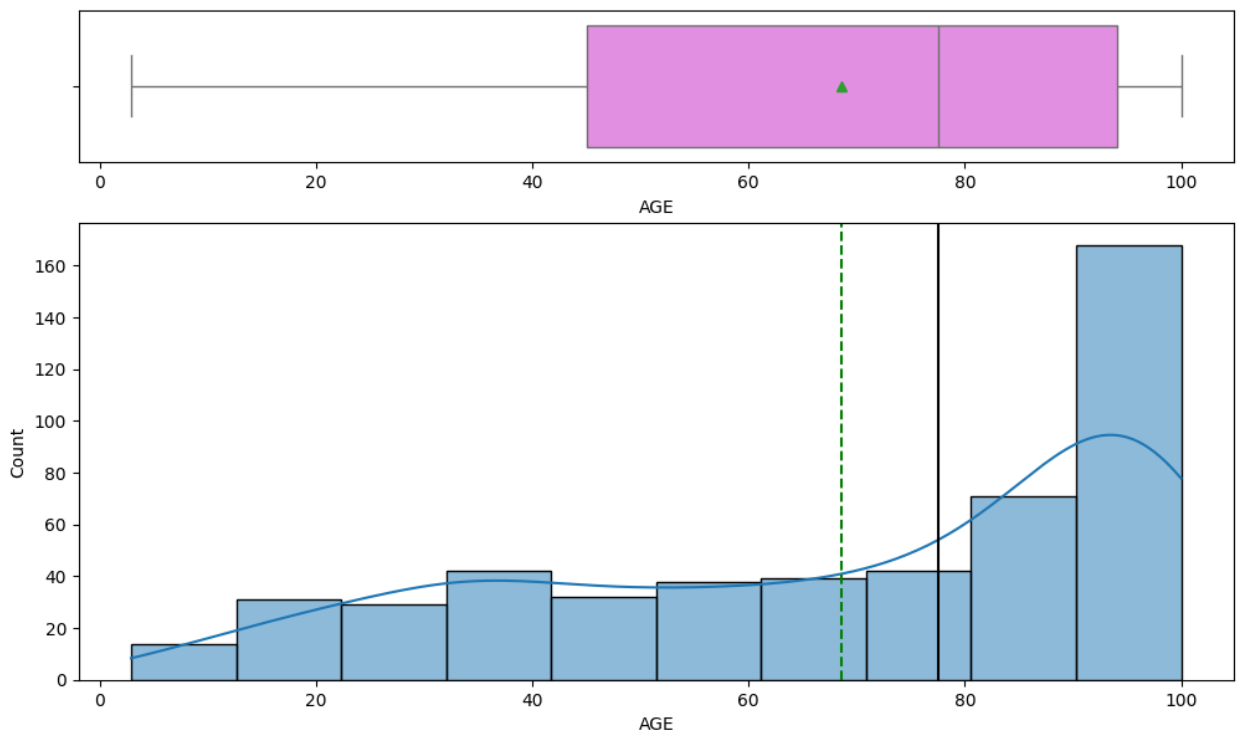
Observations and Implications on the Average number of rooms per dwelling histogram and boxplot

The histogram and boxplot of the average number of rooms per dwelling are slightly right-skewed but appear nearly normal. A significant number of homes have 5 to 7 rooms. There are fewer homes with fewer than 5 rooms and fewer homes with more than 7 rooms. Most homes are thus of moderate size. The median (denoted by a solid black line in the histogram) is 6.21, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 6.28. The boxplot shows a significant

number of outliers among homes with fewer than 5 rooms and those with more than 7 rooms. Since most homes have 5 to 7 rooms, median home values suggest middle-class neighborhoods. As fewer homes have fewer than 5 rooms and more than 7 rooms, the median home values for these homes are lower and greater, respectively.

Proportion of owner-occupied units built before 1940 (AGE) (See Appendix for function call Code Snippet)

Graph 8. Histogram and Boxplot: Proportion of owner-occupied units built before 1940 (AGE)



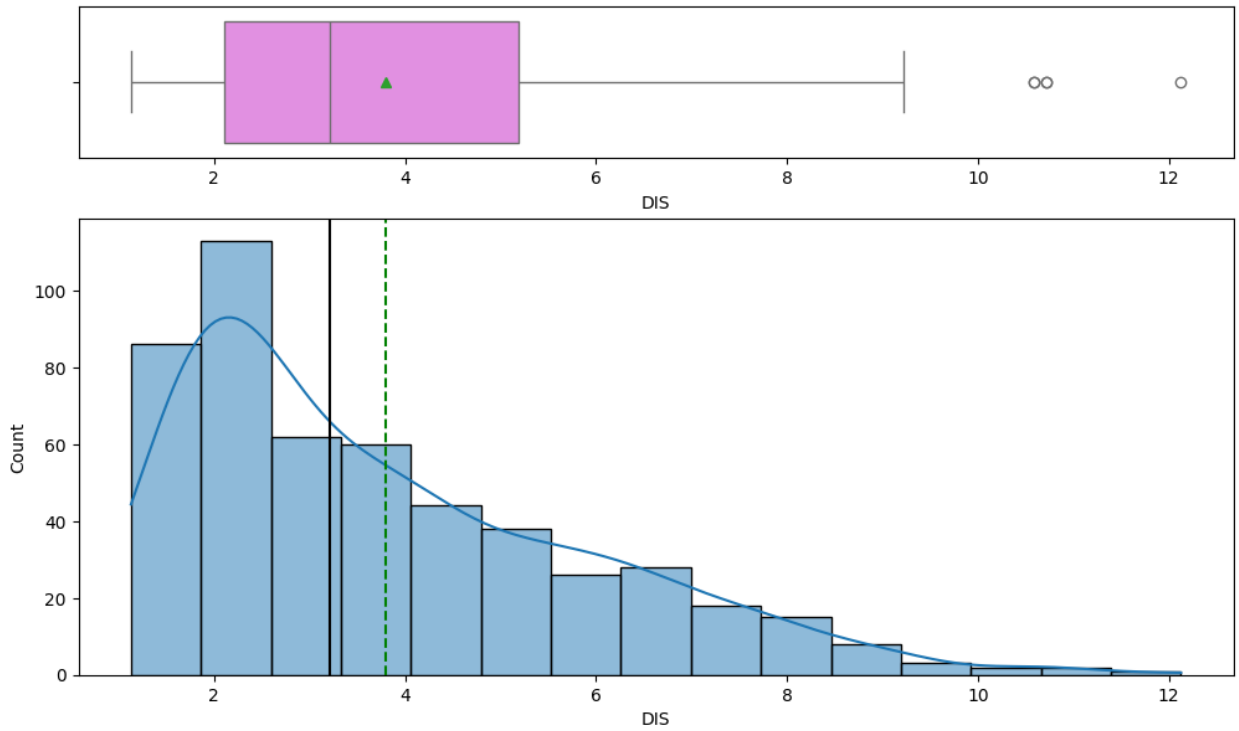
Observations and Implications on the Proportion of Owner-Occupied Units Built Before 1940

The histogram and boxplot of the proportion of owner-occupied units built before 1940 are slightly right-tailed skewed. The median (denoted by a solid black line in the histogram) is 77.50, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 68.57. A significant number of homes in towns or suburbs have AGE values between 80 and 100. There are fewer homes in towns with lower AGE values (typically lower than 40), suggesting newer homes. The median

AGE is high, at around 70. Thus, the dataset is unbalanced between older and newer homes, as most homes have higher AGE values. A shape interpretation of the AGE variable is that it shows a slight bimodality, with one cluster of newer homes and another of very old homes. As a result, towns or neighborhoods with higher AGE values have older homes that typically have lower median home values. In comparison, towns with lower AGE values have newer homes that typically have higher median home values.

Weighted distances to five Boston employment centers (DIS) (See Appendix for function call Code Snippet)

Graph 9. Histogram and Boxplot: Weighted distances to five Boston employment centers (DIS)



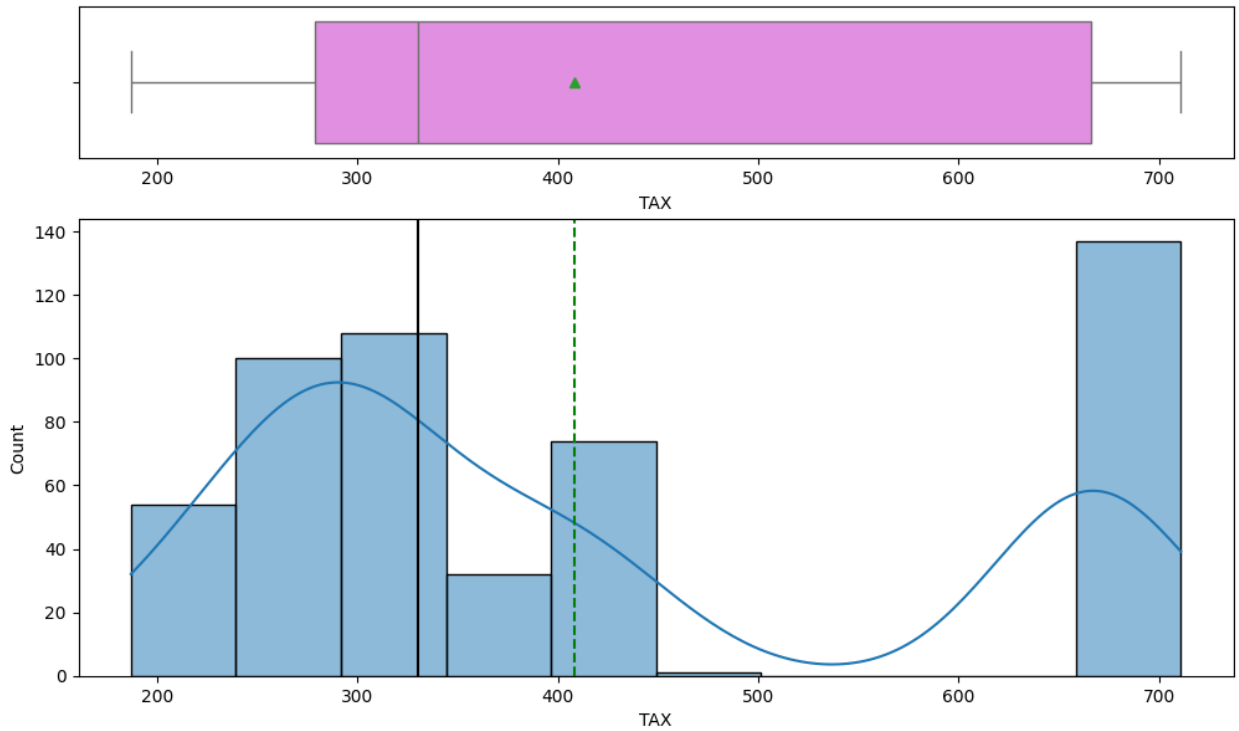
Observations and Implications on the Weighted distances to five Boston employment centers

The histogram and boxplot of the weighted distances to five Boston employment centers are significantly right-skewed. The median (denoted by a solid black line in the histogram) is 3.21, and the mean (denoted by a triangle in the boxplot

and dashed line in the histogram) is 3.80. A significant number of homes in towns have lower DIS values, approximately from 1 to 4. This suggests that most homes are closer to employment centers. In contrast, there are fewer homes in towns with higher DIS values, ranging from 5 to 12. There are some outliers with INDUS values ranging from about 11 to 12. This suggests that there are fewer homes farther from employment centers, particularly located in suburbs or exurbs. In relation to median home values, towns with lower DIS values tend to have higher median home values, reflecting their urban locations. On the other hand, towns with higher DIS values are usually associated with lower median home values, reflecting more suburban locales.

Full-value property-tax rate per $10,000 (TAX) (See Appendix for function call Code Snippet)

Graph 10. Histogram and Boxplot: Full-value property-tax rate per $10,000 (TAX)



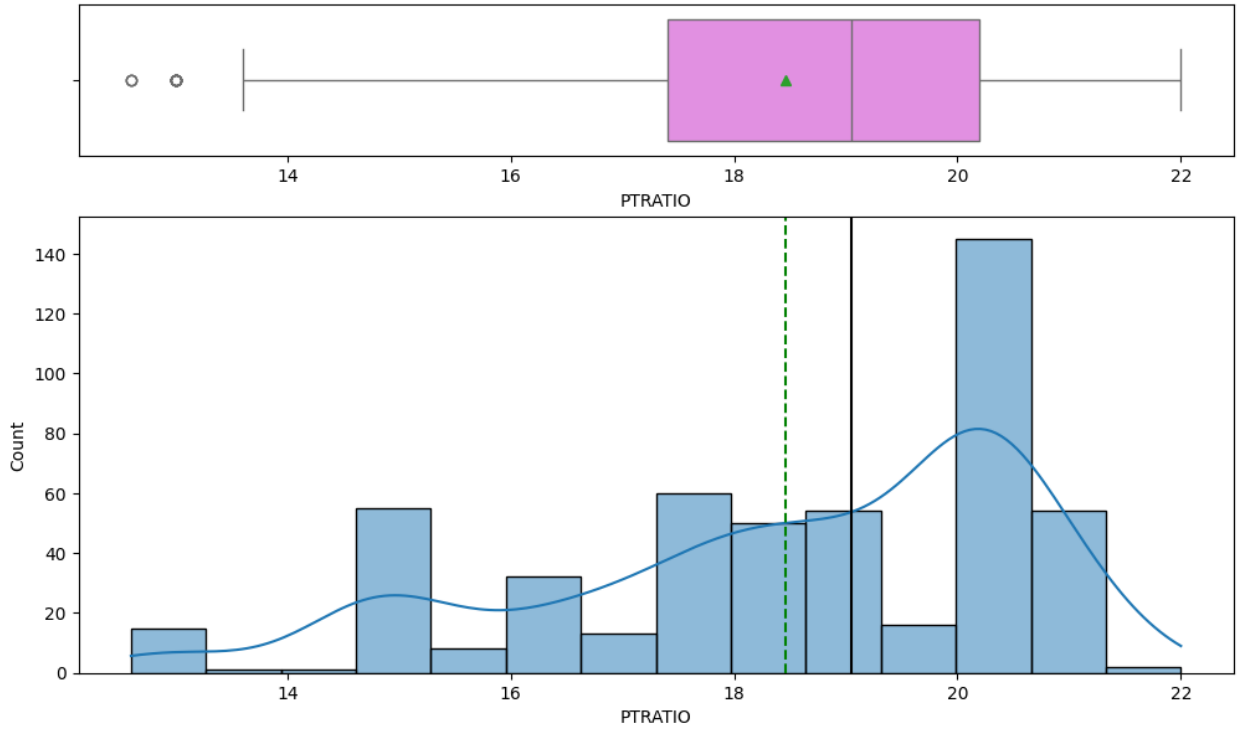
Observations and Implications on the Full-value property-tax rate per $10,000

The histogram and boxplot of the full-value property-tax rate per 10,000 are moderately right-skewed. The median (denoted by a solid black line in the histogram) is 330, and the mean (denoted by a triangle in the boxplot and dashed line in the

histogram) is 408.24. There are towns with lower- to mid-range property tax rates, ranging from 250 to 400 per $ 10,000. There is also a significant rise in higher property tax values, up to 750 per $10,000. Most towns have homes with moderate tax rates, with a median of around $350 per $10,000. These property tax values thus indicate that towns with lower property tax rates have affordable property taxes, but can sometimes be associated with higher median home values. At the other end, towns with higher property tax rates have higher municipal tax burdens and are usually associated with lower median home values.

Pupil-teacher ratio by town (PTRATIO) (See Appendix for function call Code Snippet)

Graph 11. Histogram and Boxplot: Pupil-teacher ratio by town (PTRATIO)



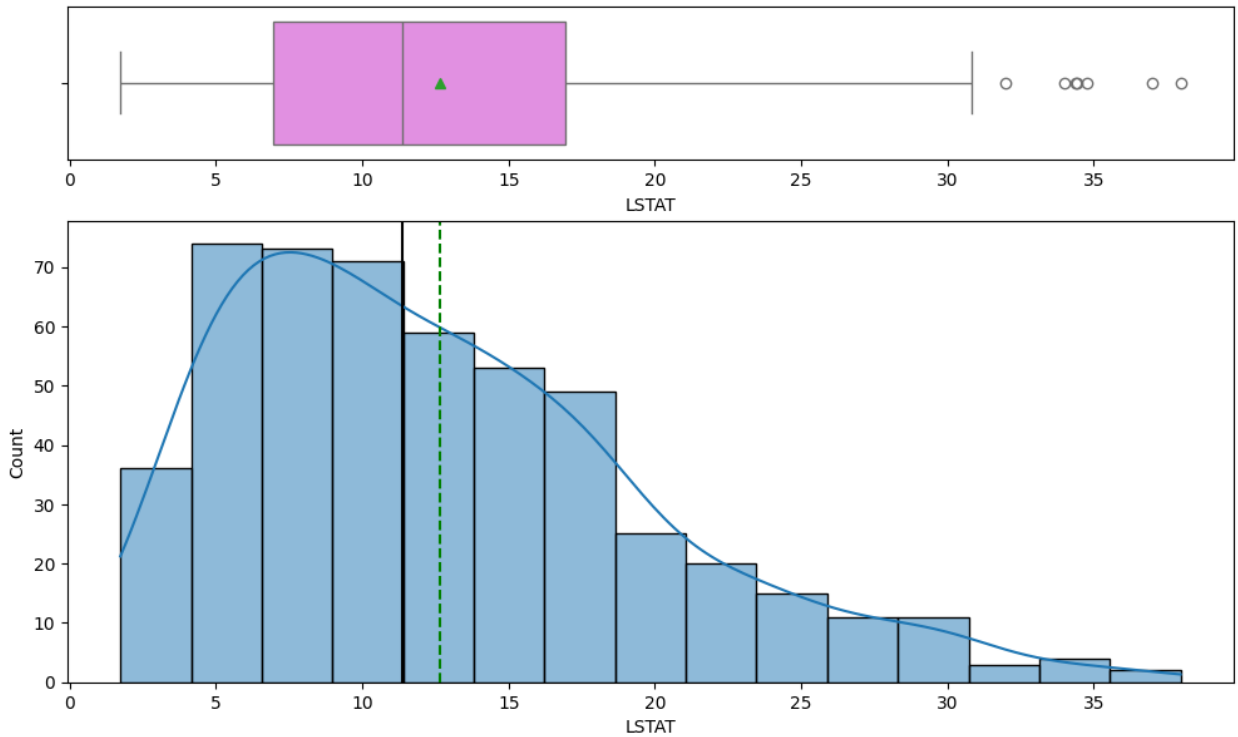
Observations and Implications on the Pupil-teacher ratio by town

The histogram and boxplot of the pupil-teacher ratio by town are fairly tight with less variability in the spread. The median (denoted by a solid black line in the histogram) is 19.05, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 18.46. The distribution is thus classified as being more

clustered. Most PTRATIO values are located within a narrow band, ranging from 16 to 22. There are two outliers, ranging from 12 to 14, representing small class sizes. The median and mean are close. In general, most towns have the same pupil-teacher ratio. With respect to median home values, the towns with higher PTRATIO values are usually associated with lower school funding and lower median home values. At the other end, towns with lower PTRATIO values are usually associated with higher school funding and higher median home values.

Percentage of lower status of the population (LSTAT) (See Appendix for function call Code Snippet)

Graph 12. Histogram and Boxplot: Percentage of lower status of the population (LSTAT)



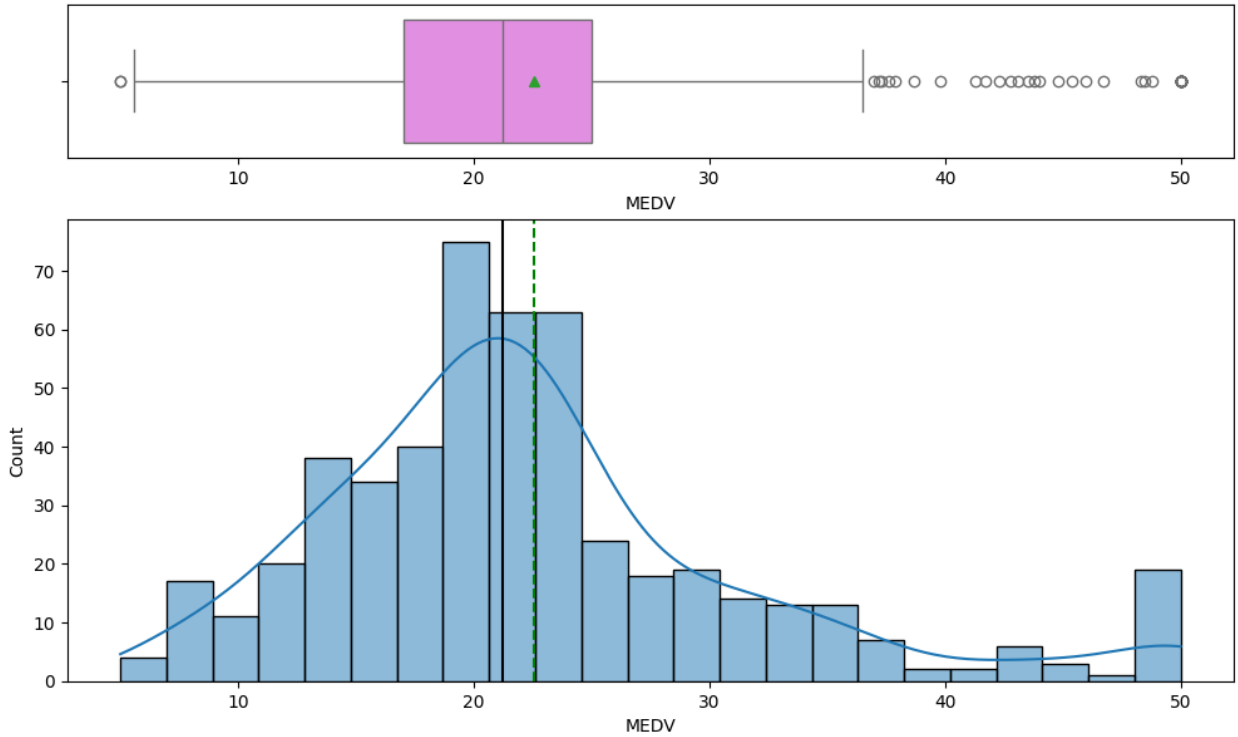
Observations and Implications on the Percentage of the lower status of the population

The histogram and boxplot of the percentage of lower-status individuals in the population are significantly right-skewed. The median (denoted by a solid black line in the histogram) is 11.36, and the mean (denoted by a triangle in the boxplot and

dashed line in the histogram) is 12.65. There is a high concentration of homes in towns with LSTAT values between 0 and 15 percent. There are also outlier LSTAT values outside the 30-40 percent range. The majority of the spread shows lower LSTAT values, ranging from about 2 to 12 percent. Thus, most towns have lower proportions of lower-status residents, and fewer towns have significantly higher proportions. In relation to median home values, towns with lower LSTAT values generally have higher median home values, and suburban areas tend to have higher education levels. On the other hand, towns with higher LSTAT values generally have lower home values, reflecting populations with greater socioeconomic disadvantages found in urban areas.

Median value of owner-occupied homes in $1,000 (MEDV) (See Appendix for function call Code Snippet)

Graph 13. Histogram and Boxplot: Median value of owner-occupied homes in $1,000 (MEDV)



Observations and Implications on the Median value of owner-occupied homes in $1,000

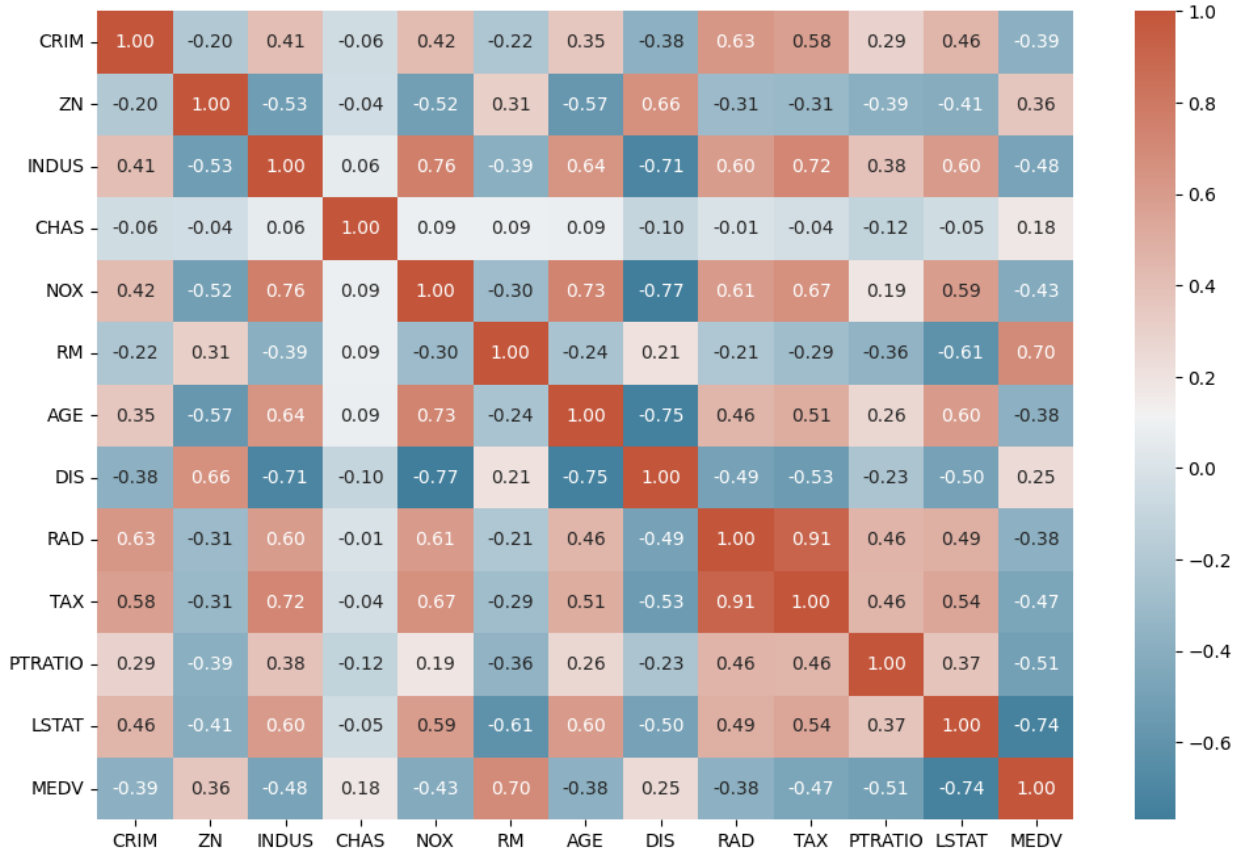
The histogram and boxplot of the median value of owner-occupied homes in $1,000 are right-skewed. The median (denoted by a solid black line in the histogram) is 21.20, and the mean (denoted by a triangle in the boxplot and dashed line in the histogram) is 22.53. There is a strong concentration of home values in Boston, ranging from $15,000 to $25,000. There is a notable number of outliers, ranging from about $ 38,000 to $50,000. There is only one outlier to the left of the whiskers, indicating a home costing less than about $5,000. Most Boston homes fall in the mid-range, whereas there are fewer that are outrageously expensive. Overall, since most towns have homes with mid-range values, this suggests middle-class neighborhoods with a moderate number of rooms, moderate LSTAT values, and moderate TAX values. At the opposite end, homes with higher values are located in low-density, low-crime areas with more rooms, lower LSTAT values, lower CRIM rates, lower NOX values, and higher ZN values.

Bivariate Analysis

Correlation among numerical variables (See Appendix for Code Snippet)

Correlation is defined (using a correlation coefficient) as how two numerical variables move together: either in a direct, positive relationship, an inverse, negative relationship, or no correlation. As in, a positive correlation suggests that both variables increase together, a negative correlation indicates that one variable increases while the other decreases, and no correlation indicates that neither variable increases or decreases. Correlation is measured using a Pearson correlation coefficient. However, a correlation does not imply causation; it only shows how variables move together.

Graph 14. Heatmap Visualization for Correlations among Categorical and Numerical Variables



Observations and Implications on the Heatmap Visualization for Correlations among Variables

The heatmap of the numerical variables in this dataset shows both strong positive and negative correlations, with varying Pearson correlation coefficients. Only variables with significant Pearson correlation coefficients will be addressed. First, there is a strong positive Pearson correlation coefficient value of 0.70 between the average number of rooms per dwelling (RM) and the median value of owner-occupied homes (MEDV). This means the more rooms a home has, the higher its median home value. Second, there is a strong negative Pearson correlation coefficient of -0.74 between the percentage of lower-status individuals in the population (LSTAT) and the median value of owner-occupied homes (MEDV). This means that the higher the percentage of the lower-status population, the lower the median home value. Third, there is a slightly strong negative Pearson correlation coefficient of -0.51 between the pupil-teacher ratio by town (PTRATIO) and the median value of owner-occupied homes (MEDV). This means that the higher the pupil-teacher ratio, the lower the median home value. Fourth, there is a slightly strong negative Pearson correlation coefficient of -0.43

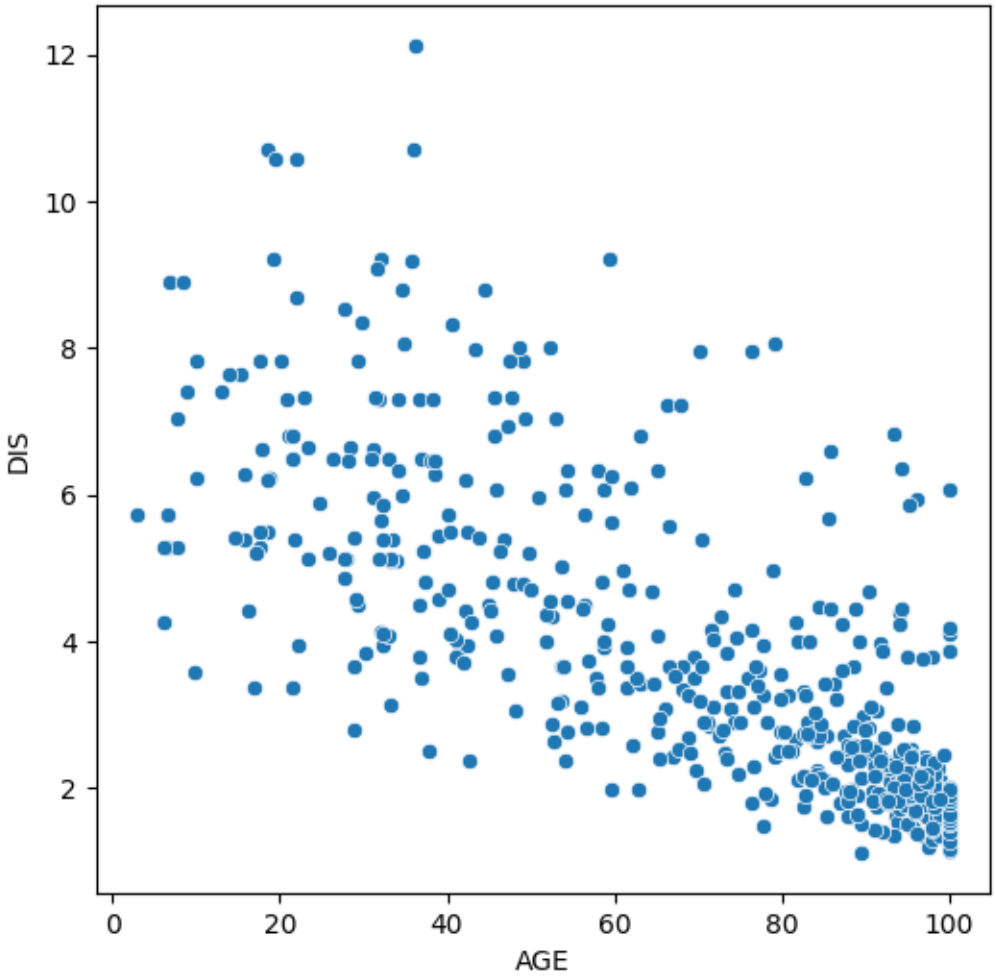
between the median nitric oxide concentration (NOX) and the median owner-occupied home value (MEDV). This means that the higher the amount of nitric oxide pollution, the lower the median home value. Fifth, there is a positive Pearson correlation coefficient of 0.25 between the weighted distances to five Boston employment centers (DIS) and the median value of owner-occupied homes (MEDV). This means that the farther from job centers, the slightly higher the median home value becomes. Sixth, there is a negative Pearson correlation coefficient of -0.39 between the per capita crime rate by town (CRIM) and the median value of owner-occupied homes (MEDV). This means that the higher the crime rates, the lower the median home value becomes. Seventh, there is a slightly significant negative Pearson correlation coefficient of -0.48 between the proportion of non-retail business acres per town (INDUS) and the median value of owner-occupied homes (MEDV). This means that the more industrialized the land, the lower the median home value becomes. Eighth, there is a positive Pearson correlation coefficient of 0.36 between the proportion of residential land zoned for lots over 25,000 square feet (ZN) and the median value of owner-occupied homes (MEDV). This means that the more large-lot zoning there is, the higher the median home value becomes. Ninth, there is a strong positive Pearson correlation coefficient of 0.76 between nitric oxide concentration (NOX) and proportion of non-retail business acres per town (INDUS). This means industrial areas tend to have higher pollution levels. Tenth, there is a strong negative Pearson correlation coefficient of -0.77 between weighted distances to five Boston employment centers (DIS) and nitric oxide concentration (NOX). This means that the farther away from an employment center, the less air pollution, indicating cleaner air. Eleventh, there is a strong negative Pearson correlation coefficient of -0.71 between weighted distances to five Boston employment centers (DIS) and proportion of non-retail businesses acres per town (INDUS). This means suburban areas have less industrial land. Twelfth, there is a relatively high positive Pearson correlation coefficient of 0.66 between the proportion of residential land zoned for lots over 25,000 square feet (ZN) and weighted distances to five Boston employment centers (DIS). This means that the farther the neighborhoods are from Boston's employment centers, the greater the tendency to have land zoned for larger, low-density residential lots. Thirteenth, there is a relatively high negative Pearson correlation coefficient of -0.57 between the proportion of residential land zoned for lots over 25,000 square feet (ZN) and the proportion of owner-occupied units built before 1940 (AGE). This means that newer homes in low-

density suburban neighborhoods are usually found in fewer older homes. In contrast, older homes are clustered in crowded urban areas where large-lot zoning is uncommon. Fourteenth, there is a very strong positive Pearson correlation coefficient of 0.91 between the index of accessibility to radial highways (RAD) and full-value property-tax rate per $10,000 (TAX). This implies that homes in towns with greater highway access generally have higher full property tax rates.

Visualizing the relationship between the independent variables (i.e., features) having significant correlations (greater than or equal to 0.7 or less than or equal to -0.7) (See Appendix for Code Snippets)

In this section, several important scatterplots are presented to visualize the significant correlations among the dataset's independent variables and the target median value of owner-occupied homes. These are variables with significant positive or negative Pearson correlation coefficients, greater than or equal to 0.7 and less than or equal to -0.7.

Graph 15. Scatterplot between Proportion of owner-occupied units built before 1940 (AGE) and Weighted distances to five Boston employment centers (DIS)

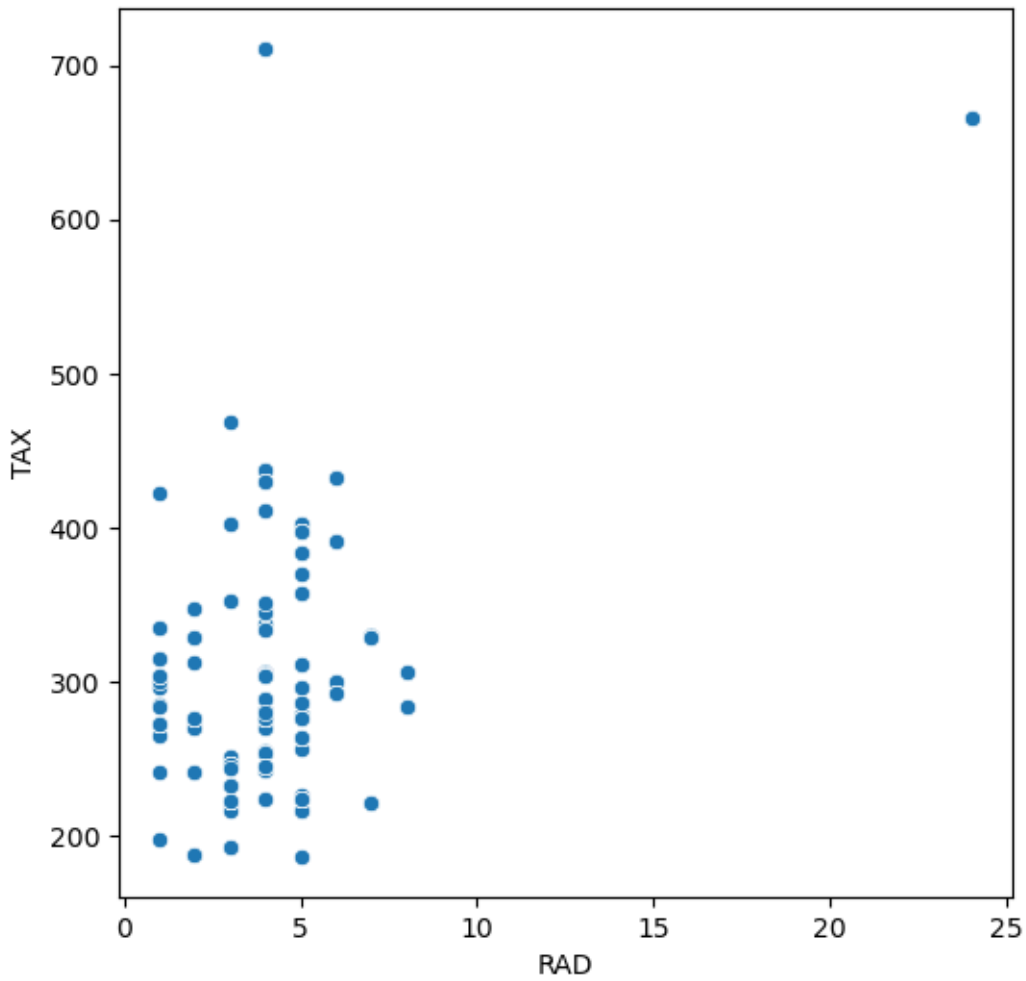


Observations and Implications between (AGE) and (DIS) Scatterplot

There is a relatively strong negative correlation (e.g., Pearson correlation coefficient of -0.75) between the proportion of owner-occupied units built before 1940 and the weighted distances to five Boston employment centers. The distance of the houses to the Boston employment centers appears to decrease moderately as the proportion of old houses increases in the town. The Boston employment centers may be located in the established towns where the proportion of owner-occupied units built before 1940 is comparatively high. In simpler terms, towns closer to Boston, with lower DIS values, have older homes, while towns farther from Boston, with higher DIS values, have newer homes. The inner city and inner-ring suburbs are closer to job centers, where homes were built earlier. There are also outer suburbs with homes that have lower AGE values and longer distances to job centers. There is a dense cluster of points with higher AGE and low DIS values. There is a spread of points with lower

AGE and higher DIS values. There are very few points with both high AGE and high DIS values.

Graph 16. Scatterplot between Index of accessibility to radial highways (RAD) and Full-value property-tax rate per $10,000 (TAX)



Observations and Implications between (RAD) and (TAX) Scatterplot

The correlation between RAD and TAX is very high, with a Pearson correlation coefficient of 0.91. No trend is visible between the two variables. Instead, the relationship is clustered, with each cluster comprising towns with the same highway accessibility index. TAX ranges from about 180 to over 700, where some RAD categories have very high TAX values. Most remaining RAD categories have moderate or low TAX values. There are only two outliers where a RAD value of 5 has a TAX

value greater than 700, and another where a RAD value of 24 has a TAX value greater than 600. A RAD value of 24 indicates extremely highway-accessible areas that are generally dense, urban, and characterized by roadway interchanges. These are areas with generally higher TAX, NOX, and CRIM values, but lower median home values. A RAD value of 1-5 indicates suburban towns, with fewer interconnected highways and less consistency with TAX values. The strong correlation might be due to outliers.

Check correlation between RAD and TAX variables again after removing outliers (See Appendix for Code Snippet)

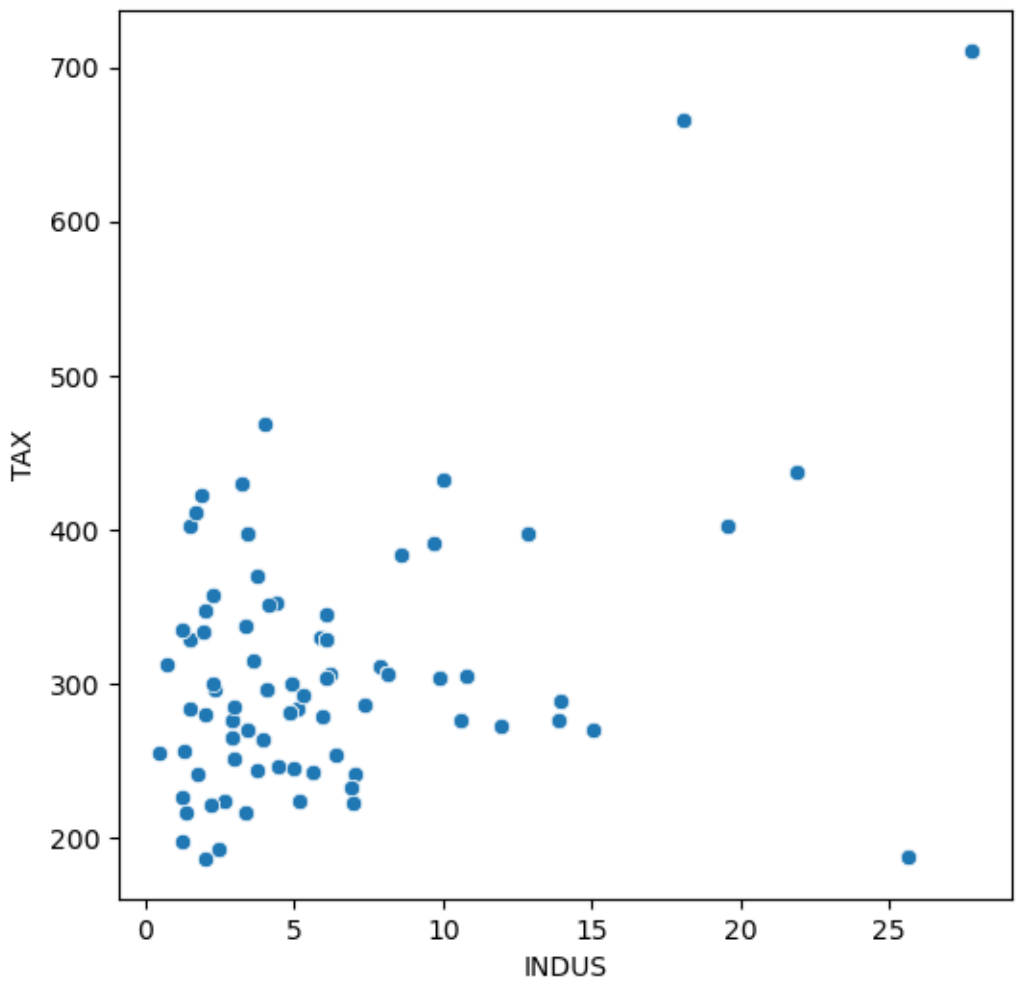
Code Output 3. Adjusted correlation between RAD and TAX variables after removing high TAX rate outliers

The correlation between TAX and RAD is 0.25.

Observations and Implications of the adjusted correlation between RAD and TAX variables after removal of high TAX rate outliers

The high correlation between TAX and RAD is due to the outliers. The tax rate for some properties might be higher for other reasons. One potential reason could be that RAD does not increase smoothly. Another potential reason is that RAD jumps between discrete categories. To clarify, TAX values vary within each RAD category, so that within-cluster group variance can lower the strength of the correlation. Once the extremely high TAX outliers are removed, the TAX distribution stabilizes, and the trend weakens. On another note, since the Pearson correlation only captures linear relationships, removing outliers introduces nonlinearity, resulting in clustered data with low correlation.

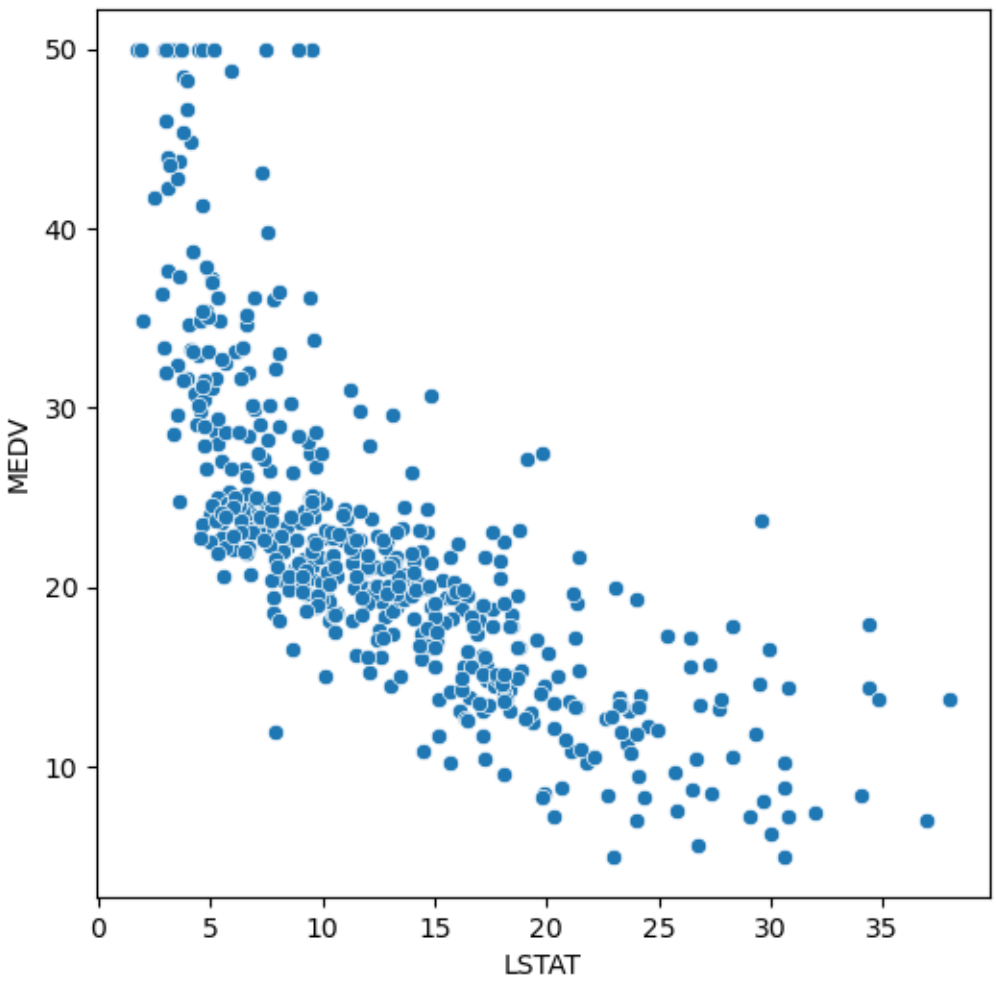
Graph 17. Scatterplot between* Proportion of non-retail business acres per town *(INDUS) and Full-value property-tax rate per $10,000 (TAX)



Observations and Implications between (INDUS) and (TAX) Scatterplot

There is a relatively strong positive correlation (e.g., a Pearson correlation coefficient of 0.72) between the proportion of non-retail business acres per town and the full-value property-tax rate per $10,000. There is a clustered cloud of points with a strong positive slope. This aligns with the fact that more industrialized areas have higher property tax rates. Most points lie within the INDUS values, ranging from 0 to 10, and the TAX values, ranging from 200 to 400. These points could be classified as a typical Boston suburb zone with moderate tax rates. The outlier points with INDUS values ranging from 15 to 25 and TAX values ranging from 500 to 700 may indicate towns with homes in high-industrialization areas and higher tax rates. Overall, the tax rate increases with the proportion of non-retail business acres per town. This may be because the INDUS and TAX variables are related to a third variable.

Graph 18. Scatterplot between Percentage of lower status of the population (LSTAT) and Median value of owner-occupied homes in $1,000 (MEDV)

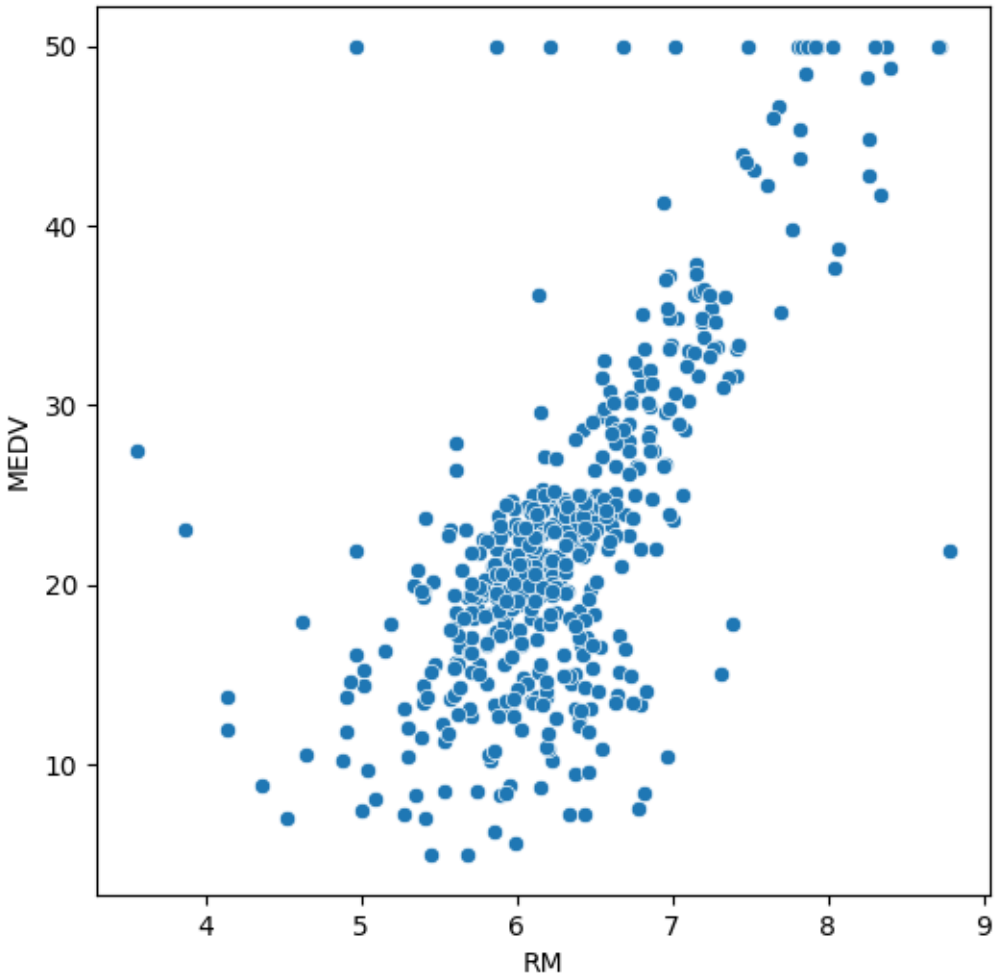


Observations and Implications between (LSTAT) and (MEDV) Scatterplot

As noted earlier, there is a strong negative Pearson correlation coefficient of -0.74 between the percentage of lower-status individuals in the population and the median value of owner-occupied homes. Again, this means that the higher the percentage of the lower-status population, the lower the median home value. In simpler terms, the price of a Boston house decreases as LSTAT increases. This could be possible because house prices are lower in areas where people of lower socioeconomic backgrounds live. There are also a few outliers in which the data appear to be capped at a MEDV value of 50 for the lowest LSTAT values. The low LSTAT values reflect higher socioeconomic status with higher Boston house price

distributions. Higher LSTAT values reflect lower socioeconomic status, with lower Boston house prices and greater variability.

Graph 19. Scatterplot between Average number of rooms per dwelling (RM) and Median value of owner-occupied homes in $1,000 (MEDV)

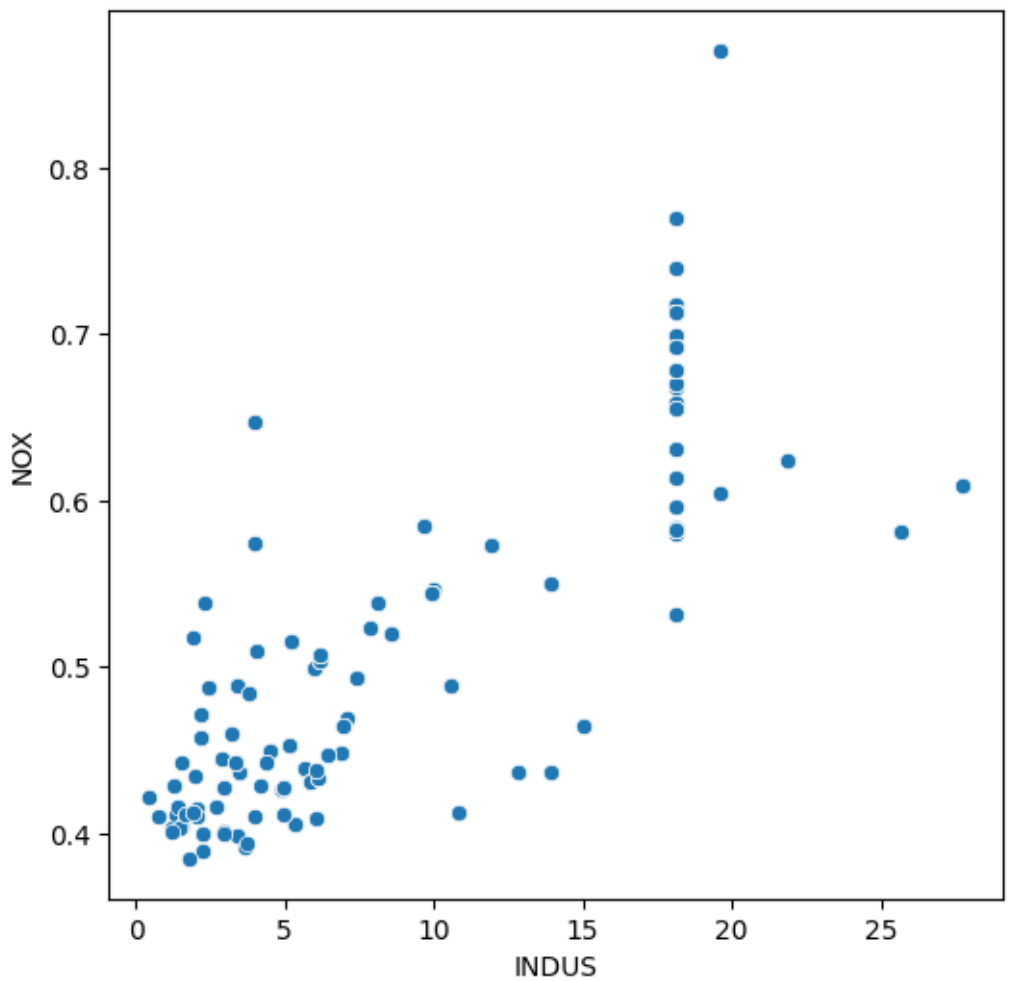


Observations and Implications between (RM) and (MEDV) Scatterplot

As previously mentioned, there is a strong positive Pearson correlation coefficient value of 0.70 between the average number of rooms per dwelling (RM) and the median value of owner-occupied homes (MEDV). The shape of the scatterplot shows that RM values in the range 4-6 are associated with slightly steady, increasing MEDV values. Also, there is a significant upward slope in RM values from about 6 to about 7.5, where larger homes are, in effect, more expensive. Additionally, there are

homes with about 8 to 9 rooms that reach capped MEDV values of about 50. As earlier stated, this means the more rooms a home has, the higher its median home value. The generalized reason could be that larger homes, with more rooms, tend to have higher home values and are most likely located in wealthier socioeconomic neighborhoods.

Graph 20. Scatterplot between Proportion of non-retail business acres per town (INDUS) and Nitric Oxide concentration (NOX)

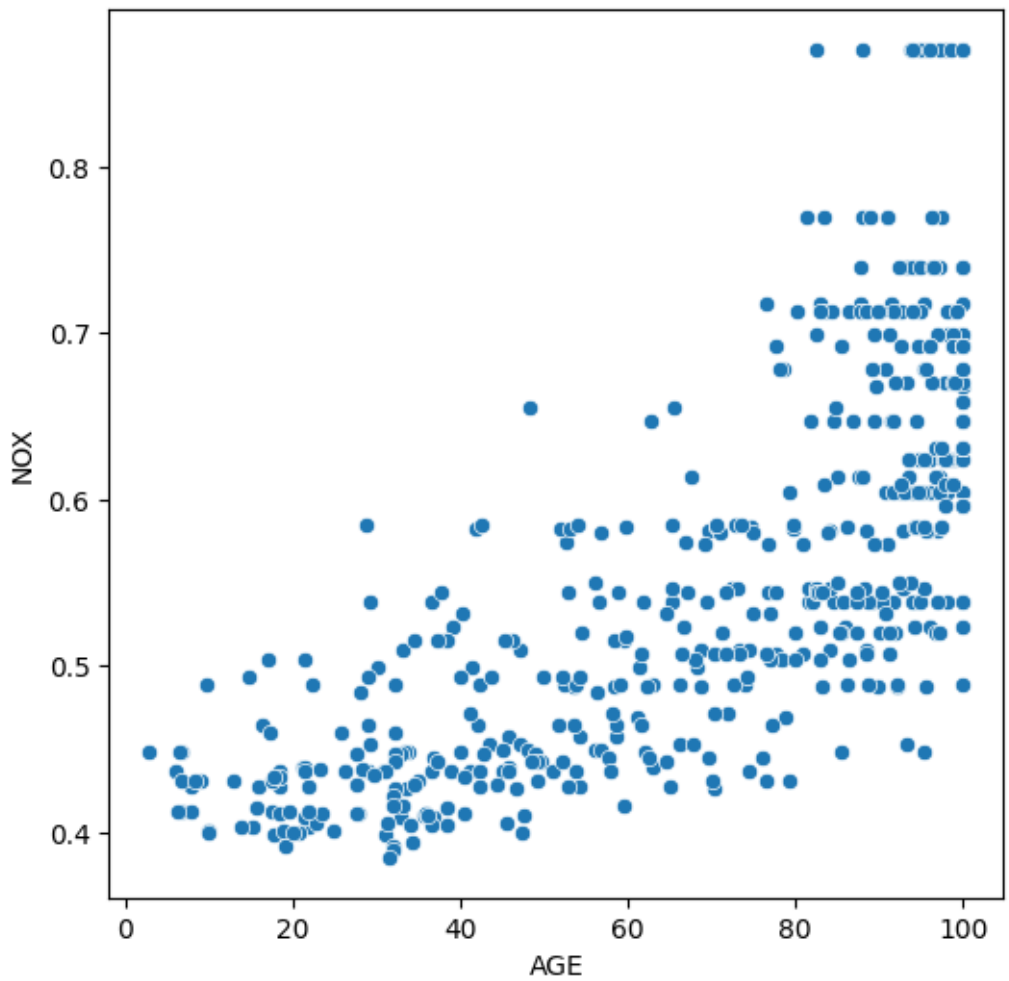


Observations and Implications between (INDUS) and (NOX) Scatterplot

As noted earlier, there is a strong positive Pearson correlation coefficient of 0.76 between nitric oxide concentration and the proportion of non-retail businesses acres per town. This means that more industrial areas tend to have higher levels of pollution. Conversely, areas with fewer industrial facilities tend to have lower pollution levels. Furthermore, homes in these areas with lower INDUS values are tightly clustered in

the scatterplot with lower NOX values. Likewise, further assumptions can be made that homes in these areas with higher INDUS values are significantly elevated in the scatterplot, with higher NOX values showing less overlap. Overall, these observations imply that areas with greater industrialization, or higher INDUS values, have higher levels of nitric oxide air pollution, and the areas with lower industrialization, or lower INDUS values, have lower levels of nitric oxide air pollution.

Graph 21. Scatterplot between Proportion of owner-occupied units built before 1940 (AGE) and Nitric Oxide concentration (NOX)

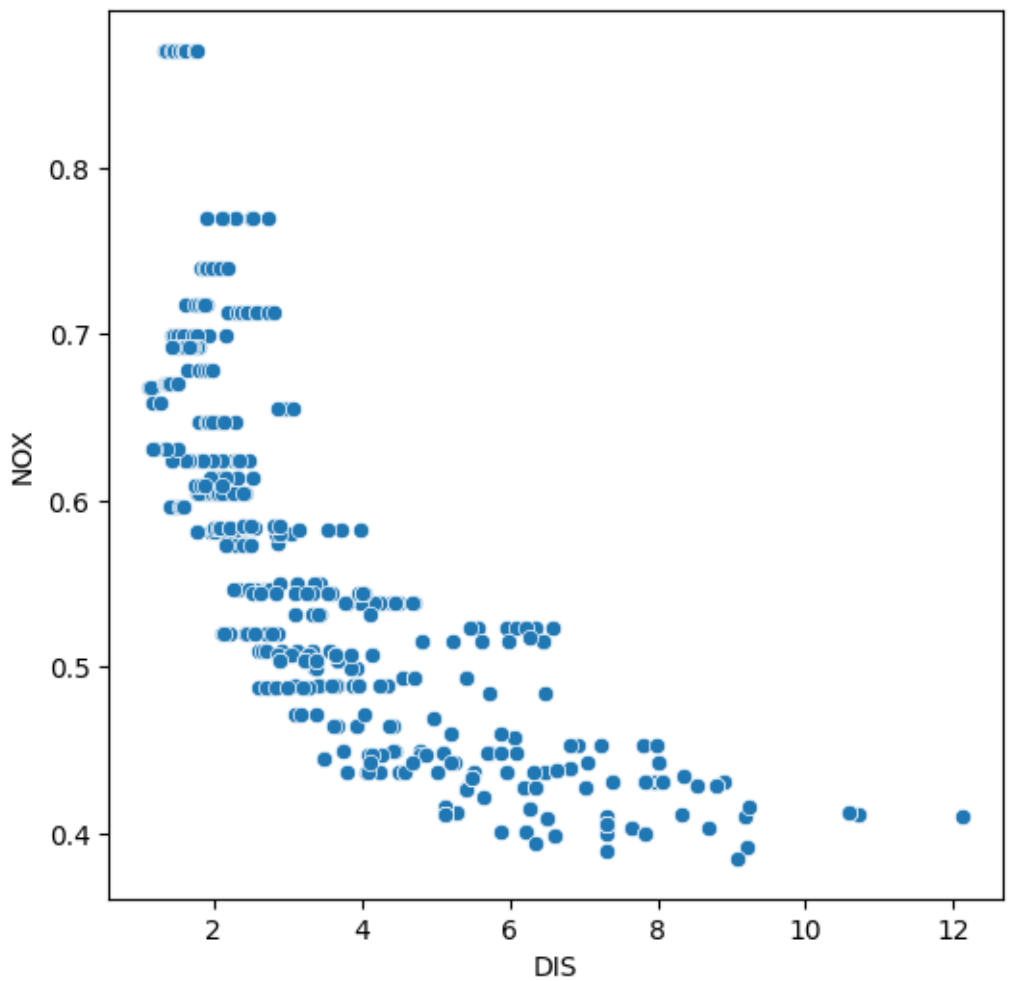


Observations and Implications between (AGE) and (NOX) Scatterplot

There is a relatively strong positive correlation (e.g., Pearson correlation coefficient of 0.73) between the proportion of owner-occupied units built before 1940 and nitric oxide concentration. As the age of a Boston home increases, nitric oxide

pollution tends to rise. Thus, the structure of the scatterplot can be described as points representing homes in towns or suburbs, with lower AGE values more tightly clustered, suggesting stabler air quality (lower NOX values) in newer developments. In contrast, points describing homes in towns or suburbs with higher AGE values are clustered together, suggesting older neighborhoods with higher levels of nitric oxide pollution (higher NOX values). Altogether, this means that older homes are located in denser urban areas, often near older road networks, and are exposed to high-traffic areas and industrial and commercial zones.

Graph 22. Scatterplot between Weighted distances to five Boston employment centers (DIS) and Nitric Oxide concentration (NOX)



Observations and Implications between (DIS) and (NOX) Scatterplot

There is a relatively strong negative correlation (e.g., Pearson correlation coefficient of -0.77) between weighted distances to five Boston employment centers and nitric oxide concentration. The closer a town or suburb is to an employment center, the higher the nitric oxide levels. On the other hand, the farther a town or suburb is from an employment center, the lower its nitric oxide levels. The shape of this relationship can be described as higher NOX levels clustering tightly near the upper range, with lower NOX levels clustering tightly at higher DIS levels. Thus, a general implication is that more centralized metropolitan areas tend to have higher NOX levels due to higher traffic volumes. Along the same lines, another implication is that the farther a town or suburb is from city metropolitan areas, the lower the NOX levels due to cleaner air.

Data Preprocessing

Outlier Handling

Based on the above univariate analysis graphs for the numerical independent variables (i.e., features) and the dependent target median value of owner-occupied homes in $1000 (MEDV) variable, no outlier values will be removed. All outlier values are true, real, and meaningful, reflecting truthful socioeconomic characteristics. Removing outliers would instead distort the distributions of these variables, introduce artificial sanitization bias, reduce model transferability, and remove significant correlation patterns. Otherwise, a separate OLS multiple linear regression model was computed and analyzed, with outliers in some variables (e.g., CRIM, ZN, DIS, PTRATIO, and LSTAT) capped. As expected, the model was significantly weaker and less potent than the other two proposed OLS multiple linear regression models, as indicated by lower R-squared values and higher Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE) values. Definitions of these performance metrics will be provided later in the report.

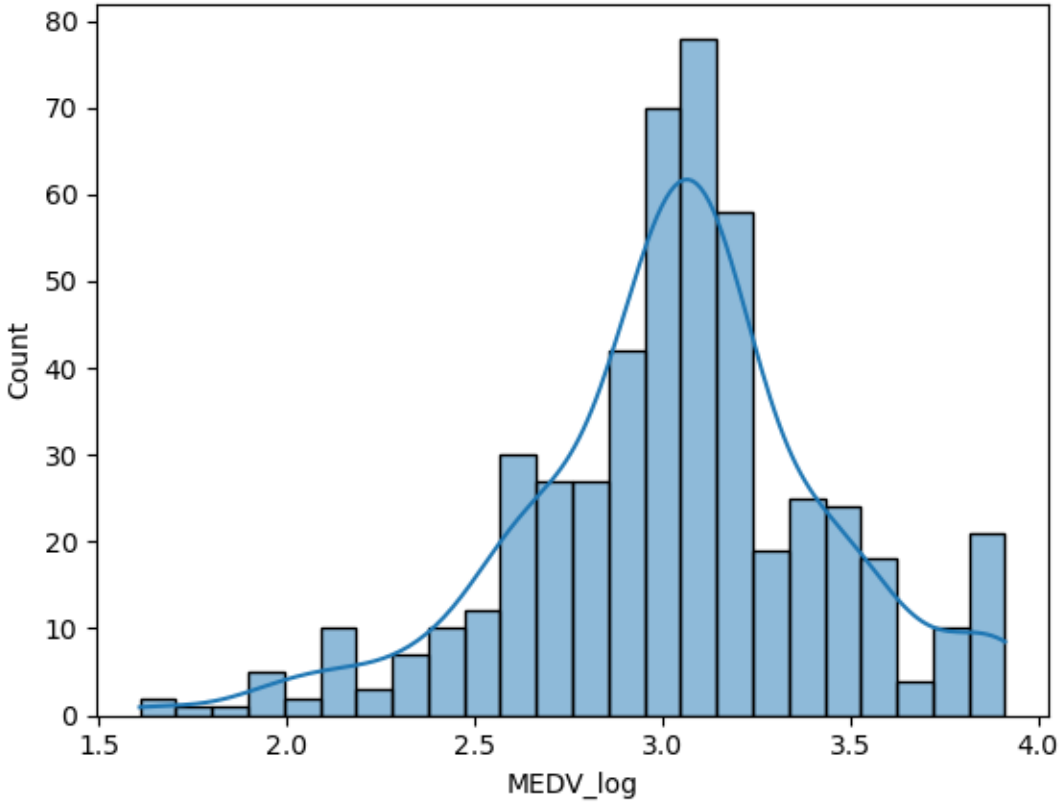
Important note about skewed independent variables with outliers

Even though the independent variables (i.e., CRIM, INDUS, DIS, and LSTAT) are highly skewed, they will be kept on their original scale and used directly in the model. Furthermore, a log transformation will not be conducted because interpreting the coefficients for each variable would become more complex.

Median value of owner-occupied homes in $1,000 log transformation (MEDV) (See Appendix for Code Snippet)

Because this is a target-dependent, skewed variable, it is necessary to apply a log transformation to the median owner-occupied home value. Before applying the log transformation, the target MEDV variable does not have stable variance and shows no linear relationships with the other independent variables. The log transformation pulls in the right tail of the original skewed distribution, making it more symmetric. This transformation allows the residuals to be normally distributed and constant, and for independent variables (or predictors) to add fixed amounts to the target. The overall rationale thus leads to improved model performance and more trustworthy multiple linear regression assumptions.

Graph 23. Log histogram transformation: Median value of owner-occupied homes in $1,000 (See Appendix for histogram plot Code Snippet)



Important note about the MEDV log variable transformation

The log-transformed variable (MEDV_log) now appears to be nearly normally distributed, with little skew. After applying this transformation, the previously mentioned aligned assumptions are met.

Categorical Variables One-Hot Encoding Conversions for Charles River dummy variable (CHAS) and Index of accessibility to radial highways (RAD) (See Appendix for Code Snippet)

Since the Charles River dummy variable is already in the preferred binary data structure, there is no need to apply one-hot encoding. However, a one-hot encoding procedure will be applied to the RAD variable to prevent the multiple linear regression model from being misled. To be specific, it would avoid the model assuming meaningful distance between distinct categories, ordinal meaning, and equal spacing between categories, as in both variables. It would also remove spurious numeric relationships, enable the model to learn independent effects for each accessibility index group, and improve overall interpretability.

Data Partitioning into features (X) and target (Y) (See Appendix for Code Snippet)

To prepare the dataset for the model, the independent variables (or features) are partitioned into an X variable, and the dependent target variable is partitioned into a Y variable. An intercept term is added to the X variable as needed for the multiple linear regression ordinary least squares (OLS) model.

Splitting the data with a ratio of train to test data (See Appendix for Code Snippet)

To set up the model evaluation, a train-test split was performed, splitting the dataset into training and test sets. The data is split into feature matrices X_train and X_test and target vectors Y_train and Y_test for training and testing. In this case, 80 percent of the data is reserved as a training set to fit the multiple linear regression model, and 20 percent is reserved as a test set to evaluate how well the model generalizes to unseen data. An 80-to-20 split was chosen, given the small size of the original dataset.

***Code output 4. Shape of the X_train dataset* (See Appendix for Code Snippet)**

(404, 20)

Observations on the Shape of the X_train Dataset

There are 404 rows and 20 columns reserved for the training dataset. Again, the original collected dataset provides a relatively very small number of observations (e.g., rows) for each column (e.g., independent variables, including the target dependent variable). Given these characteristics, it is more conservative to reserve a larger dataset for training.

***Code Output 5. Shape of the X_test dataset* (See Appendix for Code Snippet)**

(102, 20)

Observations on the Shape of the X_train Dataset

There are 102 rows and 20 columns reserved for the test dataset. The collected dataset provides a relatively very small number of observations (e.g., rows) for each column (e.g., independent variables, including the target dependent variable). Given these characteristics, it is best to use a smaller reserved dataset for the test set.

Check for Multicollinearity (See Appendix for Function Code Snippet)

Multicollinearity is a statistical phenomenon in which two or more independent variables (i.e., features or predictors) are highly correlated. To put it more simply, when two or more independent variables essentially carry the same information. Due to this phenomenon, multicollinearity can cause the individual effects of each independent variable on the target dependent variable to be isolated. It can lead to unstable coefficients, inflated variance, and inaccurate model results. Likewise, multicollinearity can make it difficult to interpret coefficients, render variables statistically insignificant, and cause numerical issues in the OLS multiple linear regression model.

The multicollinearity function was called twice to remove two independent variables having inflated Variance Inflation Factor (VIF) scores either greater than 10 or 5. Independent variables with high VIF scores indicate that they do not provide new information, increase standard errors, cause interpretational issues, or exhibit linear combinations with other variables. Within the multicollinearity function, all boolean

one-hot-encoded variables for each RAD dummy variable were converted to integer values to compute the VIF score.

***Table 6. First Multicollinearity Function result* (See Appendix for Code Snippet)**

	feature	VIF
0	const	631.374831
1	CRIM	1.743204
2	ZN	2.751841
3	INDUS	4.242444
4	CHAS	1.123667
5	NOX	4.681140
6	RM	1.993810
7	AGE	3.325811
8	DIS	4.426762
9	TAX	10.745022
10	PTRATIO	2.347767
11	LSTAT	3.046615
12	RAD_2	2.569169
13	RAD_3	3.077516
14	RAD_4	5.864641
15	RAD_5	6.405854
16	RAD_6	2.545689
17	RAD_7	2.047015
18	RAD_8	2.362674
19	RAD_24	15.627482

First Multicollinearity Function Output Table Result Observation

In this output, the RAD_24 dummy variable has a VIF score of 15.627482, which is greater than 10. This variable will first be dropped from the X_train dataset as needed (See Appendix for Code Snippet).

Table 7. Second Multicollinearity Function result (See Appendix for Code Snippet)

	feature	VIF
0	const	620.588899
1	CRIM	1.679807
2	ZN	2.665206
3	INDUS	4.088055
4	CHAS	1.118239
5	NOX	4.638782
6	RM	1.970336
7	AGE	3.310113
8	DIS	4.426292
9	TAX	6.056368
10	PTRATIO	2.317589
11	LSTAT	3.035260
12	RAD_2	2.181809
13	RAD_3	2.442372
14	RAD_4	3.410829
15	RAD_5	3.801857
16	RAD_6	1.605026
17	RAD_7	1.568032
18	RAD_8	1.738170

Second Multicollinearity Function Output Result Table Observation

In this output, the TAX independent variable has a VIF score of 6.056368, which is greater than 5. This variable will be dropped from the X_train dataset as needed (See Appendix for Code Snippet).

Table 8. Final Multicollinearity Function Output Result (See Appendix for Code Snippet)

	feature	VIF
0	const	620.124729
1	CRIM	1.636211
2	ZN	2.614195
3	INDUS	3.476529
4	CHAS	1.108493
5	NOX	4.515714
6	RM	1.957092
7	AGE	3.310112
8	DIS	4.423749
9	PTRATIO	2.201472
10	LSTAT	3.032208
11	RAD_2	1.515215
12	RAD_3	1.779928
13	RAD_4	2.136819
14	RAD_5	2.748988
15	RAD_6	1.461392
16	RAD_7	1.384117
17	RAD_8	1.506059

Final Multicollinearity Function Output Table Result Observation

In this final function output, all independent variables (or features) now have VIF scores less than 5. It is now permissible to proceed with building the first OLS multiple linear regression model after dropping the same RAD_24 and TAX variables from the X test dataset (See Appendix for the Code Snippet).

Data Types found in the X train dataset (See Appendix for Code Snippet)

Table 9. Column Variables with associated Data Types for Updated X train dataset (See Appendix for Code Snippet)

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype

```
---  ------   --------------  -----
 0   const    404 non-null    float64
 1   CRIM     404 non-null    float64
 2   ZN       404 non-null    float64
 3   INDUS    404 non-null    float64
 4   CHAS     404 non-null    int64
 5   NOX      404 non-null    float64
 6   RM       404 non-null    float64
 7   AGE      404 non-null    float64
 8   DIS      404 non-null    float64
 9   PTRATIO  404 non-null    float64
 10  LSTAT    404 non-null    float64
 11  RAD_2    404 non-null    bool
 12  RAD_3    404 non-null    bool
 13  RAD_4    404 non-null    bool
 14  RAD_5    404 non-null    bool
 15  RAD_6    404 non-null    bool
 16  RAD_7    404 non-null    bool
 17  RAD_8    404 non-null    bool
dtypes: bool(7), float64(10), int64(1)
```

Data Types found in the X train dataset (See Appendix for Code Snippet)

Table 10. Column Variables with associated Data Types for Updated X test dataset (See Appendix for Code Snippet)

Data columns (total 18 columns):

```
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   const    102 non-null    float64
 1   CRIM     102 non-null    float64
 2   ZN       102 non-null    float64
 3   INDUS    102 non-null    float64
 4   CHAS     102 non-null    int64
 5   NOX      102 non-null    float64
 6   RM       102 non-null    float64
 7   AGE      102 non-null    float64
 8   DIS      102 non-null    float64
 9   PTRATIO  102 non-null    float64
 10  LSTAT    102 non-null    float64
 11  RAD_2    102 non-null    bool
 12  RAD_3    102 non-null    bool
 13  RAD_4    102 non-null    bool
 14  RAD_5    102 non-null    bool
 15  RAD_6    102 non-null    bool
 16  RAD_7    102 non-null    bool
 17  RAD_8    102 non-null    bool
dtypes: bool(7), float64(10), int64(1)
```

Observations on Data Types found in Updated X train and X test datasets

Due to the Boolean data types found in the RAD_2, RAD_3, RAD_4, RAD_5, RAD_6, RAD_7, and RAD_8 categorical dummy variables in both the updated X train and X test datasets, these variables must undergo an integer type conversion so that

the first OLS multiple linear regression model can be built (See Appendix for Code Snippet).

Linear Regression and its Assumptions

After checking for multicollinearity and converting the RAD_2, RAD_3, RAD_4, RAD_5, RAD_6, RAD_7, and RAD_8 categorical dummy variables in both the updated X_train and X_test datasets, the first OLS multiple linear regression model (e.g., model1) is ready to be built.

Important notes about each model summary's evaluation metrics

Residuals are the calculated differences between an actual observed value and the value predicted by a model. They are critical in determining how well a model best fits the data. An R-squared value ranges from 0 to 1 and indicates the proportion of the variation in a dependent variable explained by the independent variables (predictors) in a multiple linear regression model. The closer the R-squared value is to 1, the more of the outcome's variance the model captures. An adjusted R-squared value is a modified alternative to the R-squared statistic that measures how much of the variation in the dependent variable is explained after penalizing for the number of predictors, and it ranges from 0 to 1. It improves on regular R-squared by correcting the fact that R-squared typically increases after adding more variables, even when they are useless. A p-value for an independent variable is defined as the measurement of the probability that its relationship with the target dependent variable is due to random chance, assuming the null hypothesis is true. A low p-value (less than 0.05) indicates that the variable has a statistically significant effect on the dependent variable. As a result, the null hypothesis can be rejected, and the variable is considered a significant predictor. Conversely, a high p-value (greater than 0.05) indicates that the variable has no statistically significant effect on the dependent variable. As a result, the null hypothesis cannot be rejected, and the variable is not considered a significant predictor.

Important notes on Model performance on evaluation metrics for X train and X test data

Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Difference (MAPE) are evaluation metrics used for OLS multiple linear regression models. RMSE measures the typical size of a model's prediction errors in the same units as the target variable. It begins by squaring the residuals, averaging them, and taking the square root of the result. MAE measures the average size of a model's prediction errors, where it treats every error equally and maintains the result in the same units as the target variable. It uses absolute residuals, obtained by subtracting the actual values from the predicted values and taking the absolute value of the difference. MAPE measures the average percentage difference between actual and predicted values, as used in residuals, but converts them to percentage errors. It divides each error by the actual value, converts the result to a percentage, and then averages those percentages across all observations. Again, the R-squared value indicates how much of the variation in a dependent variable is explained by the independent variables (predictors) in a multiple linear regression model. The closer the R-squared value is to 1, the more of the outcome's variance the model captures. For optimal model performance, RMSE, MAE, and MAPE evaluation metrics should be as low as possible.

Table 11. First Model (model1) OLS multiple linear regression Model Summary (See Appendix for Code Snippet)

OLS Regression Results

Dep. Variable:	MEDV_log	**R-squared:**	0.780
Model:	OLS	**Adj. R-squared:**	0.770
Method:	Least Squares	**F-statistic:**	80.57
Date:	Fri, 27 Feb 2026	**Prob (F-statistic):**	1.98e-115
Time:	22:22:41	**Log-Likelihood:**	98.645
No. Observations:	404	**AIC:**	-161.3
Df Residuals:	386	**BIC:**	-89.26
Df Model:	17		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	4.2989	0.240	17.893	0.000	3.827	4.771
CRIM	-0.0098	0.001	-7.238	0.000	-0.012	-0.007
ZN	0.0013	0.001	1.953	0.052	-8.76e-06	0.003
INDUS	0.0002	0.003	0.063	0.950	-0.005	0.005
CHAS	0.1049	0.038	2.790	0.006	0.031	0.179
NOX	-0.8391	0.176	-4.779	0.000	-1.184	-0.494
RM	0.0718	0.020	3.663	0.000	0.033	0.110
AGE	0.0004	0.001	0.651	0.516	-0.001	0.002
DIS	-0.0523	0.010	-5.461	0.000	-0.071	-0.033
PTRATIO	-0.0365	0.007	-5.521	0.000	-0.049	-0.023
LSTAT	-0.0297	0.002	-12.750	0.000	-0.034	-0.025
RAD_2	-0.0126	0.052	-0.240	0.810	-0.115	0.090
RAD_3	0.0573	0.048	1.202	0.230	-0.036	0.151
RAD_4	-0.0466	0.035	-1.348	0.178	-0.115	0.021
RAD_5	-0.0225	0.038	-0.589	0.556	-0.097	0.053
RAD_6	-0.0874	0.053	-1.664	0.097	-0.191	0.016
RAD_7	0.0665	0.060	1.108	0.269	-0.052	0.185
RAD_8	0.0107	0.057	0.186	0.852	-0.102	0.124

Omnibus:	45.563	**Durbin-Watson:**	1.948
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	179.647
Skew:	0.395	**Prob(JB):**	9.78e-40
Kurtosis:	6.170	**Cond. No.**	2.16e+03

Observations and Implications on First Model (model1) OLS multiple linear regression Model Summary

The OLS multiple linear regression model summary for model 1 (i.e., model1) has 404 observations, 386 residuals, an R-squared value of 0.780, and an Adjusted R-

squared value of 0.770. The R-squared and Adjusted R-squared values here indicate a strong fit to the real-world data and show that nearly all independent variables (predictors) contribute meaningful information. The log transformation of the dependent variable, MEDV, makes the effects more proportional and stabilizes the variance. Although given the previously stated p-value distinction, it is important to note the p-values for each independent variable (feature), including the constant term. Here, the constant term and the variables CRIM, CHAS, NOX, RM, DIS, PTRATIO, and LSTAT each have corresponding p-values of 0.000, 0.000, 0.006, 0.000, 0.000, 0.000, and 0.000, respectively. Because each p-value is less than 0.05, these independent variables are considered statistically significant and will be retained for a second OLS multiple linear regression model. Instead, the ZN, INDUS, AGE, RAD_2, RAD_3, RAD_4, RAD_5, RAD_6, RAD_7, and RAD_8 variables each have the corresponding p-values: 0.052, 0.950, 0.516, 0.810, 0.230, 0.178, 0.556, 0.097, 0.269, and 0.852. Because each p-value is greater than 0.05, these independent variables will be considered for removal, and a second model will be fitted without them (See Appendix for Code Snippet). The following implications for this first model suggest that the model captures how socioeconomic status, nitric oxide pollution, crime, school quality, and housing size predominantly determine Boston home prices across suburbs and towns. Similarly, geographic proximity variables, such as distance to different Boston job hubs and homes near the Charles River tract, also determine Boston home prices across suburbs and towns.

Check First Model (model1) performance (See Appendix for Code Snippet for model performance function)

Table 12. Checking model1 performance on X train and X test data

	Data	RMSE	MAE	MAPE	r2
0	Train	0.189549	0.134353	4.671885	0.718177
1	Test	0.205496	0.155262	5.441155	0.615417

Observations and Implications on Checking model1 performance on X train and X test data

Here, the training and test datasets perform relatively well. The training dataset has an RMSE of 0.189, an MAE of 0.134353, an MAPE of 4.671885, and an R-squared of 0.718177. The testing dataset has an RMSE of 0.205496, an MAE of 0.155262, an

MAPE of 5.441155, and an R-squared of 0.615417. The RMSE test error is slightly higher, indicating a normal generalization gap. The test error for MAE is increased modestly on unseen data. The MAPE predictions on the test dataset show strong accuracy. The R-squared value for the test data indicates lower variance. Taken together, the model generalizes fairly well, the R-squared evaluation metric on the test dataset is solid, the RMSE and MAE evaluation metrics confirm mild tail errors, but still indicate model stability and less outlier dominance, and the MAPE evaluation metric shows that, on average, the model's predictions are off by only about 5% of the time. These evaluation metrics imply that the model is excellent and applicable to real-world housing data, and that there is no overfitting between the training and test sets.

Check Second Model (model2) performance (See Appendix for Code Snippet for model performance function)

Table 13. Model2 OLS multiple linear regression Model Summary (See Appendix for Code Snippet)

OLS Regression Results

Dep. Variable:	MEDV_log	**R-squared:**	0.771
Model:	OLS	**Adj. R-squared:**	0.767
Method:	Least Squares	**F-statistic:**	190.4
Date:	Fri, 27 Feb 2026	**Prob (F-statistic):**	1.60e-122
Time:	22:23:48	**Log-Likelihood:**	90.399
No. Observations:	404	**AIC:**	-164.8
Df Residuals:	396	**BIC:**	-132.8
Df Model:	7		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	4.1437	0.205	20.221	0.000	3.741	4.547
CRIM	-0.0087	0.001	-7.036	0.000	-0.011	-0.006
CHAS	0.1070	0.037	2.897	0.004	0.034	0.180
NOX	-0.7785	0.143	-5.463	0.000	-1.059	-0.498
RM	0.0911	0.018	4.929	0.000	0.055	0.127
DIS	-0.0418	0.007	-5.748	0.000	-0.056	-0.027
PTRATIO	-0.0376	0.005	-7.413	0.000	-0.048	-0.028
LSTAT	-0.0290	0.002	-13.535	0.000	-0.033	-0.025

Omnibus:	56.955	**Durbin-Watson:**	1.988
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	213.493
Skew:	0.563	**Prob(JB):**	4.37e-47
Kurtosis:	6.379	**Cond. No.**	562.

Observations and Implications on Second Model (model2) OLS multiple linear regression Model Summary

As previously mentioned, the ZN, INDUS, AGE, RAD_2, RAD_3, RAD_4, RAD_5, RAD_6, RAD_7, and RAD_8 independent variables (features) are considered for removal from the first model (model1) because their p-values exceed 0.05. Following their removal, the OLS multiple linear regression model summary for model 2 (i.e., model2) has 404 observations, 386 residuals, an R-squared value of 0.771, and an Adjusted R-squared value of 0.767. Again, the R-squared and Adjusted R-squared values here indicate a strong fit to the real-world data and show that nearly all independent variables (predictors) contribute meaningful information. The remaining CRIM, CHAS, NOX, RM, DIS, PTRATIO, and LSTAT independent variables are highly significant due to having p-values less than 0.05. The CRIM independent variable indicates that higher crime rates are associated with lower home prices. The CHAS

independent variable indicates that homes adjacent to the Charles River are priced higher. The NOX independent variable indicates that nitric oxide pollution significantly reduces home prices. The RM independent variable implies that the more rooms a home has, the higher its price. The DIS independent variable implies that the farther a home is from employment centers, the higher its housing price is. The PTRATIO independent variable implies that the higher the student-teacher ratio, the lower the housing pricing is. The LSTAT independent variable implies that the higher the percentage of lower-status individuals in the population, the lower housing prices are.

Check Second Model (model2) performance (See Appendix for Code Snippet for model performance function)

Table 14. Checking model2 performance on X train and X test data

	Data	RMSE	MAE	MAPE	r2
0	Train	0.193457	0.138025	4.782150	0.702947
1	Test	0.205768	0.157446	5.515539	0.616398

Observations and Implications on Checking model2 performance on X train and X test data

Here, the training and test datasets perform relatively well. The training dataset has an RMSE of 0.193457, an MAE of 0.138025, a MAPE of 4.782150, and an R-squared of 0.702947. The testing dataset has an RMSE of 0.205768, a MAE of 0.157446, a MAPE of 4.782150, and an R-squared of 0.616398. The RMSE error on the test dataset is lower than that on the training dataset, indicating no overfitting, stable residual variance, and appropriate model complexity. The MAE on the test dataset is higher than on the training dataset, indicating that extreme values are not distorting the model. The MAPE error is around 5-6 percent, indicating acceptable performance for housing data. The R-squared value on the testing dataset indicates less variance in the test data. Taken together, this model also generalizes fairly well; again, it is not overfitting, interpretability has improved, MAPE suggests feasible accuracy, and the residual makeup is robust. Due to the second model having a slightly higher R-squared value on the testing dataset, no statistically weak variables (e.g., ZN, INDUS, AGE, RAD_2, RAD_3, RAD_4, RAD_5, RAD_6, RAD_7, and RAD_8), and no distortion of signs and magnitudes of the statistically strong variables, the second model (model2) will be chosen as the preferred model. Moreover, the second model

strengthened, stabilized, and recovered the effects for the CRIM, CHAS, NOX, RM, DIS, PTRATIO, and LSTAT variables. Most importantly, the second model also has higher overall interpretability, stability, and alignment with economic intuition on the housing pricing.

Checking for the four multiple linear regression assumptions is an important technique to ensure the reliability and validity of the model

After proceeding with the second model (model2), it is critical to ensure that this OLS multiple linear regression model meets four assumptions, as each assumption safeguards its trustworthiness. First, the residuals in the model should have a mean of 0. Second, the residuals should be homoscedastic, meaning they are symmetrically distributed around the regression line. A Goldfeld-Quandt Test will be used to assess whether the residuals are homoscedastic (i.e., not heteroscedastic) in the model by verifying that the p-value exceeds 0.05, and that the F-statistic score is close to 1. In other words, there is a failure in the rejection of the null hypothesis. Third, the independent variables (features) in the model must be linearly related to the target MEDV variable. For this assumption, the residuals are plotted and do not exhibit a strong pattern, and are randomly and uniformly scattered on the x-axis. Fourth, the residuals in the model must be normally distributed, or, in other words, the error terms are normally distributed. To test this assumption, a Q-Q plot and a histogram of the residuals are used.

Mean of residuals should be 0

***Code Output 6.* (See Appendix for Code Snippet)**

np.float64(-4.8052870813355164e-15)

Observations and Implications on the Mean of residuals should be 0 output

The model's mean residuals are very close to 0. The number in decimal form is -0.00000000000000480528708133551 64. This means that this first assumption is met. This further implies that the model is not over- or under-predicting. Also, this implies that the residual diagnostics are functioning normally.

Check for Homoscedasticity

Code Output 7. (See Appendix for Code Snippet)

[('F statistic', np.float64(1.0880964374565942)),

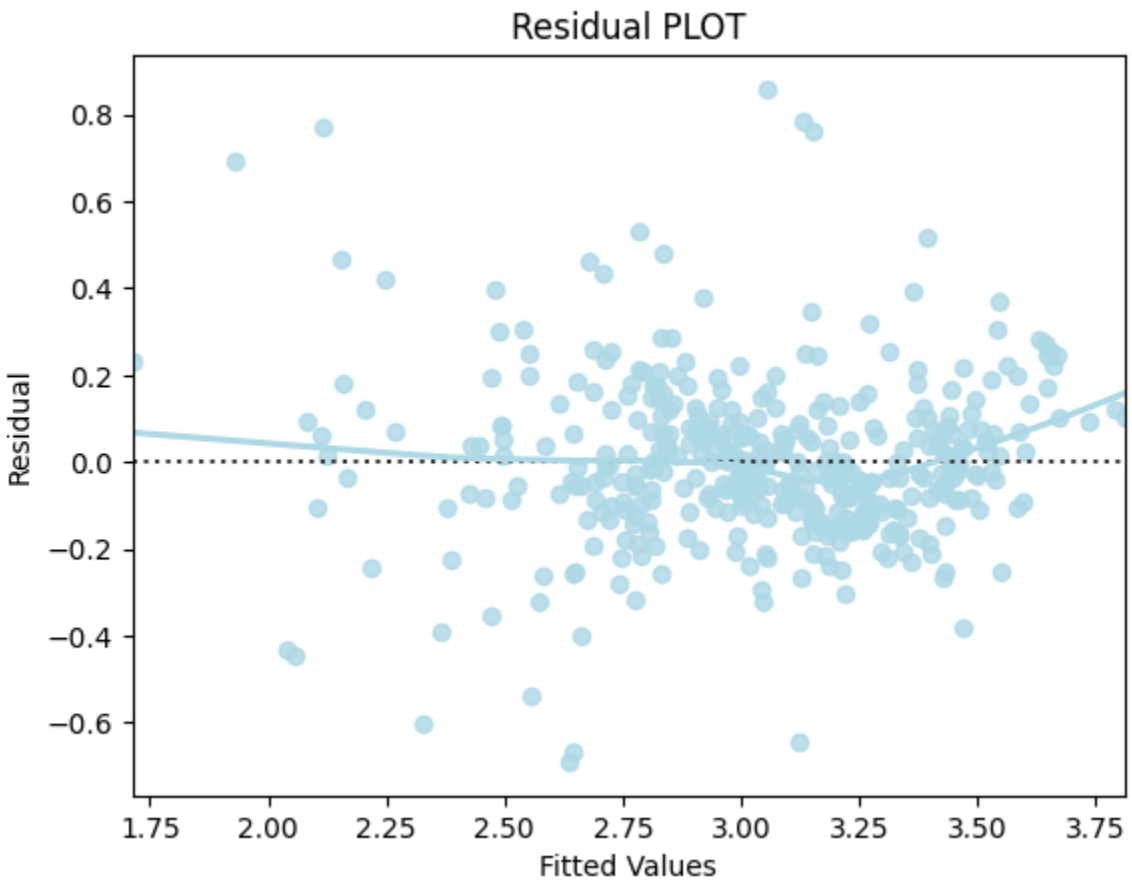
('p-value', np.float64(0.2785544698968608))]

Observations and Implications on Checking for Homoscedasticity

Here, the p-value based on the code output is 0.2785544698968608, and the F-statistic value is 1.0880964374565942. The p-value is greater than 0.05, indicating that the null hypothesis cannot be rejected. The F-statistic is close to 1, indicating that the residual variances are very similar. The interpretation of these results is that the model residuals are homoscedastic. These results imply that the p-values and F-statistics are reliable and, when interpreted together, provide no statistical evidence of heteroscedasticity in the residuals.

Linearity of variables

Graph 24. Residual Plot (See Appendix for Code Snippet)

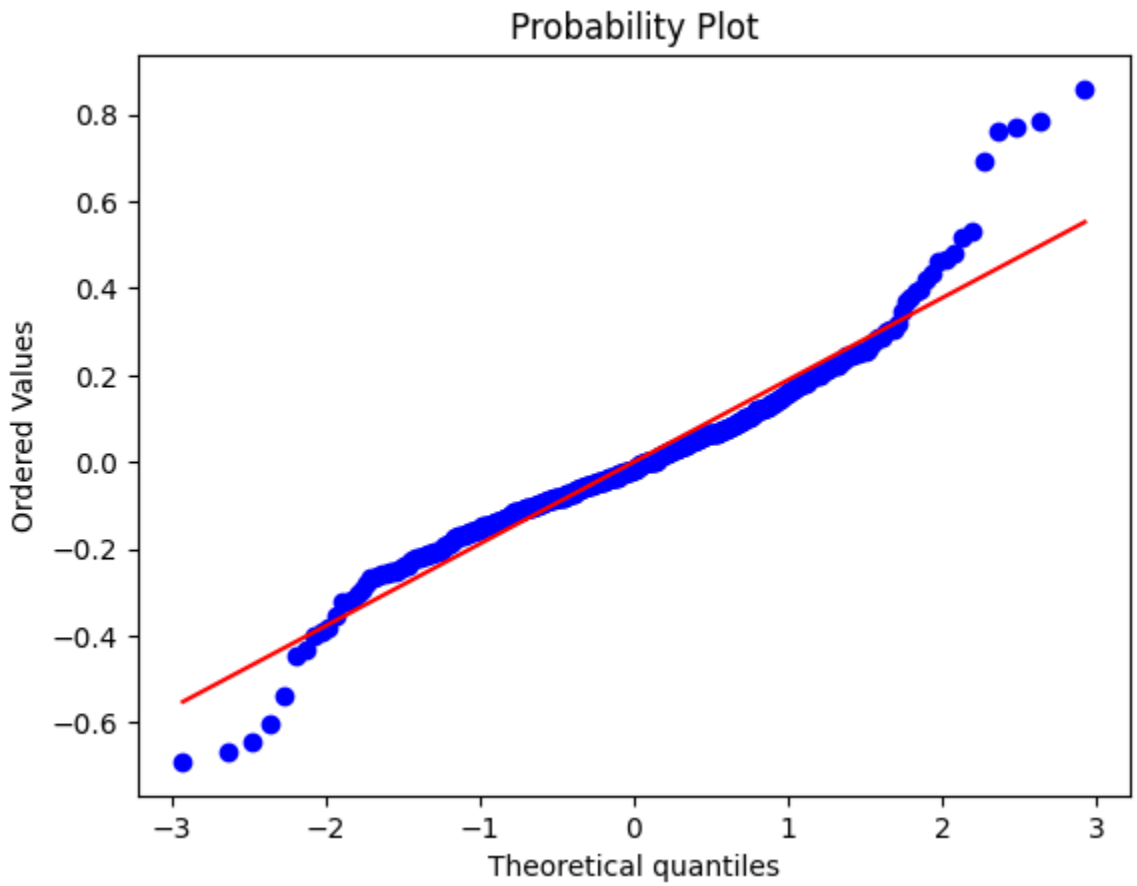


Observations and Implications on Residual Plot

In this residual plot, the residuals and the fitted values do not form a noticeable strong pattern. They are randomly and uniformly scattered on the x-axis. Thus, the condition for the linearity of the independent variables in the model is satisfied. Other implications can be drawn, such as constant variance, where standard errors and p-values are trustworthy, error independence, and no strong indication of omitted nonlinear terms.

Normality of error terms

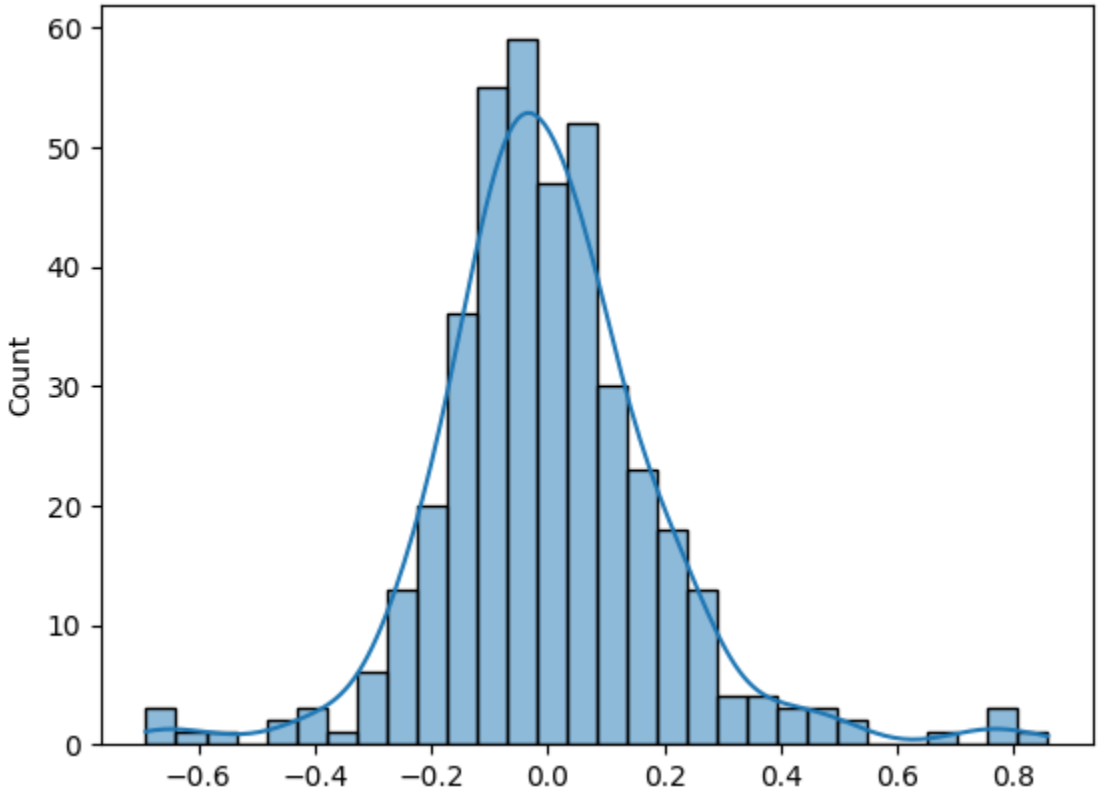
Graph 25. Residuals Q-Q plot (See Appendix for Code Snippet)



Observations and Implications on the Residuals Q-Q Plot

In this Q-Q residuals plot, the theoretical red straight reference line captures most of the ordered residuals (represented as blue) in the plot. In other words, the points stay close to the theoretical red line throughout the majority of the distribution, and do not deviate much away, except at the tails of the blue graphed line. These plots indicate that the residuals follow a normal distribution, with no significant anomalies, such as skewness or outliers.

Graph 26. Residual Histogram (See Appendix for Code Snippet)



Observations and Implications on Residuals Histogram

In this residuals histogram plot, the residuals follow a strict bell-shaped curve. The bell-shaped curve peaks near zero, resulting in a symmetric relationship on both sides of zero. Most residuals are near zero, indicating that the model's predictions are overall accurate. There is no evidence of asymmetry or extreme outliers, suggesting no signs of non-normality, such as tails or skewness. These findings are aligned with the Q-Q plot.

Technique of applying Cross-validation to improve the model

Cross-validation is used to improve the model's performance. The rationale behind this technique is to prevent overfitting, provide a more reliable estimate of performance, corroborate model complexity, detect data issues, and use the data more effectively. In this case, the performance metrics, R-squared and mean squared error, are used in a cross-validation framework. Mean squared error (MSE) is a measure of the average squared prediction error in a multiple linear regression model, calculated

as the average of the squared differences between predicted and actual values. Similar to RMSE, MAE, and MAPE, MSE should be as low as possible.

Code Output. 8 (See Appendix for Code Snippet)

RSquared: 0.729 (+/- 0.239)

Mean Squared Error: 0.040 (+/- 0.023)

Observations and Implications on Cross-validation Output

After applying cross-validation, the R-squared value is 0.729, with a variability of 0.239 (positive) and 0.239 (negative), and the mean squared error is 0.040, with a variability of 0.023 (positive) and 0.023 (negative). The R-squared value indicates that, across all folds, the model now explains approximately 73% of the variance in the target MEDV variable. This indicates a strong level of explanatory power for this OLS multiple linear regression model. The variability of this R-squared score indicates that some folds, or subsets of the training and testing datasets, consistently perform well across all portions of the dataset. The mean squared error value implies that the model's average squared error is small, which is in alignment with the strong R-squared value. The variability of this mean squared error is that some folds, or subsets of the training and testing datasets, are moderately accurate.

Second model (model2) coefficients

The coefficients for the second model (model2) are obtained. The independent variables, or features, for this model are CRIM, CHAS, NOX, RM, DIS, PTRATIO, and LSTAT.

Table 15. Second Model (model2) coefficients (See Appendix for Code Snippet)

	Feature	Coefs
0	const	4.143698
1	CRIM	-0.008742
2	CHAS	0.106978
3	NOX	-0.778458
4	RM	0.091108
5	DIS	-0.041784
6	PTRATIO	-0.037619
7	LSTAT	-0.029007

Equation 1. Model2 multiple linear regression equation

$\log($ **MEDV** $) = 4.143698 - 0.008742*$ **CRIM** $+0.106978*$ **CHAS** $-0.778458*$ **NOX** $+0.091108*$ **RM** $-0.041784*$ **DIS -0.037619 * PTRATIO -0.029007 * LSTAT**

Observations and Implications on the Second model (model2) coefficients table and equation

CRIM has a coefficient value of -0.008742. The negative coefficient implies that higher crime rates lead to lower home values, with an estimated 0.87 percentage-point decrease in the predicted home price. CHAS has a coefficient value of 0.106978. The positive value implies that homes near the Charles River have about a 10.7 percent premium included within the total cost. NOX has a coefficient value of -0.778458. The negative value implies that nitric oxide pollution strongly reduces home value, with an estimated 7.8% decrease in the predicted home price. RM has a coefficient value of 0.091108. The positive value indicates that the more rooms a home has, the greater its value, with an approximate 9.1 percent increase in the predicted price. DIS has a coefficient value of -0.041784. The negative value indicates that the farther a home is from an employment center, the lower its value. There is also an approximately 4.2 percent decrease in home value associated with this variable. PTRATIO has a coefficient value of -0.037619. The negative value indicates that higher pupil-teacher ratios (i.e., poor-performing overcrowded schools) are associated with lower home values. There is also an approximately 3.8 percent decrease in home

value associated with this variable. LSTAT has a coefficient value of -0.029007. The negative value indicates that the higher the percentage of lower-status populations, the lower the home value. There is also an approximately 2.9 percent decrease in home value associated with this variable. Taken together, the strongest independent variables in the model are NOX, RM, and LSTAT; the moderate-performing independent variables are PTRATIO, DIS, and CHAS; and the weakest-performing independent variable is CRIM.

Important insights from the multiple linear regression equation model

Based on the developed multiple linear regression model, specific housing marketing strategies, urban planning and environmental policy planning strategies, education and community investment strategies, transportation and accessibility planning strategies, real estate development and investment strategies, and policy evaluation and scenario planning strategies can be developed. To start with, based on the RM, NOX, and LSTAT independent variables in the model in relation to a housing marketing strategy, there should be efforts that highlight home renovation, identification of homes in undervalued neighborhoods, and efforts to deter home buyers from overpaying for homes in areas with high nitric oxide pollution or a high percentage of lower-status population. Secondly, based on the NOX and LSTAT independent variables in the model in relation to urban planning and environmental policy, home construction planners should aim at reducing pollution in areas where the nitric oxide levels are prominent, allocate green buffers, zoning alterations, and industrial relocations, and invest resources in neighborhoods with a higher percentage of lower-status populations. Thirdly, based on the PTRATIO independent variable in the model in relation to education and community investment, community representatives should prioritize funding decisions for schools to have smaller class sizes, implement school improvements in areas with lower home values, and inform stakeholders (e.g., home buyers, property investors, and housebuilders) of the necessity of how educational investment assets influence economic gains in the housing market. Fourthly, based on the DIS independent variable in the model in relation to transportation and accessibility planning, planners should validate transit expansions or increase the number of employment hubs, and recognize communities that are not close to employment hubs. Fifthly, real estate investors and developers can use the proposed multiple linear regression model to identify the independent

variables (e.g., CHAS and RM) that yield the highest investment returns. In addition, real-estate investors and developers can evaluate whether a home's value aligns with its predicted value and, based on neighborhood trends, estimate the projected long-term rise in home values. Lastly, important local Boston government officials can estimate the driving impact of evolving policy changes, most notably reductions in nitric oxide pollution levels, run several “what-if” scenarios for home development, and communicate housing pricing expectations based on quantitative analysis to community stakeholders.

Conclusion

Actionable Insights

Real estate businesses and affiliated stakeholders (e.g., home buyers, property investors, and housebuilders) can apply the following actionable insights. To begin with, real estate businesses and stakeholders can allocate resources to improving housing quality, including prioritizing home renovations and building larger, multi-room houses. Secondly, real estate businesses and affiliated stakeholders (e.g., home buyers, property investors, and housebuilders) can support environmental and pollution-reduction efforts by identifying neighborhoods with high levels of nitric oxide pollution and implementing eco-friendly buffers, such as tree planting and traffic rerouting. Thirdly, real estate businesses and affiliated stakeholders (e.g., home buyers, property investors, and housebuilders) can enhance public transit connectivity to employment hubs for homes in neighborhoods farther from employment centers. Fourthly, real estate businesses and affiliated stakeholders (e.g., home buyers, property investors, and housebuilders) can apportion funds to lower the student-teacher ratios in poor-performing schools, thereby increasing home values. Fifthly, real estate businesses and affiliated stakeholders (e.g., home buyers, property investors, and housebuilders) can promote community programs that enhance socioeconomic conditions across towns and suburbs, thereby increasing home property values. Sixthly, real estate businesses and affiliated stakeholders (e.g., home buyers, property investors, and housebuilders) can implement crime-reduction and neighborhood-safety measures, such as identifying crime hotspots, improving street lighting, and implementing community safety initiatives, to improve home values. Lastly, real estate businesses and affiliated stakeholders (e.g., home buyers, property investors, and

housebuilders) can leverage the preferred-housing-amenity effect, in which homes are valued more when near the Charles River, by constructing recreational areas near bodies of water to boost home values.

Significance of the techniques used

The regression techniques used to develop the multiple linear regression model provide a quantitative, evidence-based foundation for a better understanding of the predictor variables. The use of regression diagnostics and cross-validation techniques ensured that the insights were statistically sound and operationally reliable. The modeling process translated complex business dynamics into clear, easy-to-understand relationships, enabling stakeholders (e.g., home buyers, property investors, and housebuilders) to apply the model to their needs. The residual and assumption criteria validated that the model generalizes across both training and testing datasets, increasing overall confidence for stakeholders (e.g., home buyers, property investors, and housebuilders), as it performs well in real-world real estate conditions. The cross-validation technique ensures that the model performs well on unseen data, corroborating forecasting stability and supporting continuous planning. The feature significance testing process helped identify the variables with the strongest influence, enabling real estate stakeholders (e.g., home buyers, property investors, and housebuilders) to allocate resources accordingly. The multicollinearity-checking technique ensured that the model's estimates were fixed and interpretable, while avoiding inaccurate conclusions regarding the important drivers. The performance metrics, such as R-squared and RMSE, provided transparency into the model's performance and accuracy, helping stakeholders (e.g., home buyers, property investors, and housebuilders) better understand its durability and constraints. The strategic implications of these techniques are that the model provides a strong foundation for scenario planning, a quantification of marginal effects analysis for each predictor or independent variable, a methodological rigor enforced in its creation, and an aggregation of statistical testing and predictive evaluation that positions the model.

Finalized the multiple linear regression model coefficients interpretation

As previously stated, the coefficients are -0.008742 for CRIM, 0.106978 for CHAS, -0.778458 for NOX, 0.091108 for RM, -0.041784 for DIS, -0.037619 for PTRATIO, and -0.029007 for LSTAT. These coefficient values, obtained for each

independent variable (or predictor), have the following interpretations. Again, the Boston home values are mostly determined by structural features found across homes (i.e., RM), environmental quality from air pollution (i.e., NOX), socioeconomic demographical contexts (i.e., LSTAT and PTRATIO), home localities (i.e., CHAS and DIS), and crime rates (i.e., CRIM). To emphasize, homes with more rooms and have adjacency locales near the Charles River are worth more, homes located in areas with higher nitric oxide pollution levels are worth less, homes with higher percentage of lower status of the population are worth less, homes located near poor performing schools with larger pupil-teacher ratios are worth less, homes with less accessibility to employment centers are worth less, and homes in areas with higher crime rates are worth less. Overall, the interpretations of the coefficient values from the multiple linear regression model indicate that buyers place considerable value on homes in clean environments, highly prominent community attributes, and superb-quality housing features. In contrast, buyers would not place a high value on homes with higher levels of nitric oxide pollution, greater distance from employment hubs, and poor school quality. Nevertheless, the crime rate mainly plays a modest role compared to the environmental and socioeconomic predictors. By applying a log transformation to the target MEDV variable, real estate stakeholders can use percentage-based insights to quantify how cumulative changes in each predictor (independent variable) translate into expected home prices across Boston suburbs or towns.

Proposal of specific actions that the business can take based on the multiple linear regression model analysis

Based on the multiple linear regression model analysis, six actionable insights for real estate businesses have been identified. First, real estate businesses can use the pricing strategy for Boston homes across various suburbs or towns by setting data-driven listing prices for properties based on measurable neighborhood attributes, such as RM, LSTAT, NOX, and PTRATIO. These real estate businesses can thus classify lower-valued homes whose current market prices are below the predicted price. Similarly, these real estate businesses can highlight higher-value homes whose asking prices exceed the predicted price to improve negotiation leverage. Second, real-estate businesses can make better investment and renovation decisions by using variables, such as CHAS and RM, to prioritize which property features drive home values. These efforts include home renovations, such as adding rooms and improving interior quality.

Other efforts include reducing investment in home renovations for properties in areas with unfavorable conditions, as measured by variables such as NOX and LSTAT. Most importantly, real estate businesses can use the model's predicted outputs to calculate the Return on Investment for housing upgrades, as a means to pilot overall investment strategies. Third, real estate businesses can strategically identify which areas are suitable for specific site selections. Specifically, real estate developers and investors can use the model to simulate future value scenarios under different land zoning changes or enhancements to the framework. In the same manner, real estate developers and investors can use the model to identify areas with homes of prominent negative values based on independent variables with negative driving coefficients (e.g., CRIM, NOX, DIS, PTRATIO, and LSTAT). Fourth, real estate business stakeholders, such as government and urban-planning representatives, can use the model to evaluate independent variables with negative driving coefficients (e.g., CRIM, NOX, DIS, PTRATIO, and LSTAT) by incorporating interventions, such as improving school funding and lowering nitric oxide pollution levels. Fifth, real estate businesses can use the model to group potential purchasers by preferences, such as room space, school quality, or proximity to employment centers. Sixth, real estate businesses can use the model to advertise properties with the strongest positive independent variables (e.g., CHAS and RM), and to predict which neighborhoods align with purchasers' preferences.

How the analysis translate to business success

The multiple linear regression model analysis contributes to the success of real estate businesses through the following factors. First, the multiple linear regression model provides quantitative evidence by minimizing financial risk and improving the accuracy of predicted home prices. Second, the multiple linear regression model allows real estate businesses to identify which independent variables, or predictors, are most critical, while avoiding misallocation of time and money to housing improvements or asset misestimates. Third, the multiple linear regression model enables real estate businesses to gain a stronger competitive advantage by providing accurate home prices to clients and buyers and offering data-driven, evidence-based explanations for different home values. Fourth, the multiple linear regression model indicates that real estate businesses should adopt long-term strategic planning measures, including portfolio diversification, development planning, policy advocacy, and the scheduling of

market decisions. Fifth, the multiple linear regression model allows real estate businesses to enhance communication with all types of stakeholders (e.g., home buyers, property investors, and housebuilders), as each separate coefficient in the seven significant predictors facilitates clear explanations for reasons detailing why particular homes in neighborhoods have higher values, why certain suburbs or towns have home values that increase over a shorter period than expected, and which targeted interventions can improve home values. The overall multiple linear regression model analysis directly enhances profitability and reduces risk for real estate businesses while applying data-driven decision-making across real estate investments and urban planning. In summary, the key takeaways from the multiple linear regression model are the following: increased home size and homes located near the Charles River have increased home values, nitric oxide pollution levels highly negate home values, socioeconomic factors frame buyer demand where buyers pay more for homes located near high performing schools and higher socioeconomic demographics, close connectedness to employment centers matter where homes farther away from employment centers lower their value, and the lowering of crime rates increase home values.

Appendix (List of Code Snippets)

Code Snippet 1. Output the first 5 rows found in the Dataset

```
# Return the first 5 rows
df.head()
```

Code Snippet 2. Output the last 5 rows found in the Dataset

```
# Return the last 5 rows
df.tail()
```

Code Snippet 3. Shape of the Dataset

```
# Check the shape of the dataset
df.shape
```

Code Snippet 4. Check the Data Types of the Dataset

```
# Check the data types of the different columns in the dataset
df.info()
```

Code Snippet 5. Check for missing values for each column in the Dataset

```
# Check for missing values for each column in the dataset
df.isnull().sum()
```

Code Snippet 6. Check for duplicate values in the Dataset

```
# Check for duplicate entries in the dataset
df.duplicated().sum()
```

Code Snippet 7. Obtain the summary Descriptive Statistics for all numerical and categorical variables

```
# Get the summary statistics of the all numerical and categorical variables
df.describe().round(2).T
```

Code Snippet 8. Labeled barplot function for categorical non-numerical variables

```python
# Function to create labeled barplots for categorical non-numerical variables

def labeled_barplot(data, feature, perc = False, n = None):
    """
    Barplot with percentage at the top

    data: dataframe
    feature: dataframe column
    perc: whether to display percentages instead of count (default is False)
    n: displays the top n category levels (default is None, i.e., display all levels)
    """

    total = len(data[feature])            # Length of the column
    count = data[feature].nunique()
    if n is None:
        plt.figure(figsize = (count + 1, 5))
    else:
        plt.figure(figsize = (n + 1, 5))

    plt.xticks(rotation = 90, fontsize = 15)
    ax = sns.countplot(
        data = data,
        x = feature,
        palette = "Paired",
        order = data[feature].value_counts().index[:n].sort_values(),
    )

    for p in ax.patches:
        if perc == True:
            label = "{:.1f}%".format(
                100 * p.get_height() / total
            )                             # Percentage of each class of the category
        else:
            label = p.get_height()        # Count of each level of the category

        x = p.get_x() + p.get_width() / 2  # Width of the plot
        y = p.get_height()                 # Height of the plot

        ax.annotate(
            label,
            (x, y),
            ha = "center",
            va = "center",
            size = 12,
            xytext = (0, 5),
            textcoords = "offset points",
        )                                 # Annotate the percentage

    plt.show()                            # Show the plot
```

Code Snippet 9. Boxplot and Histogram function for numerical variables

```python
# Function to plot a boxplot and a histogram along the same scale for numerical variables

def histogram_boxplot(data, feature, figsize = (12, 7), kde = False, bins = None):
    """
    Boxplot and histogram combined

    data: dataframe
    feature: dataframe column
    figsize: size of figure (default (12,7))
    kde: whether to show density curve (default False)
    bins: number of bins for histogram (default None)
    """
    f2, (ax_box2, ax_hist2) = plt.subplots(
        nrows = 2,      # Number of rows of the subplot grid = 2
        sharex = False,  # x-axis will not be shared among all subplots
        gridspec_kw = {"height_ratios": (0.25, 0.75)},
        figsize = figsize,
    )                   # Creating the 2 subplots
    sns.boxplot(
        data = data, x = feature, ax = ax_box2, showmeans = True, color = "violet"
    )                   # Boxplot will be created and a star will indicate the mean value of the column
    sns.histplot(
        data = data, x = feature, kde = True, ax = ax_hist2, bins = bins, palette = "winter"
    ) if bins else sns.histplot(
        data = data, x = feature, kde = True, ax = ax_hist2
    )                   # For histogram
    ax_hist2.axvline(
        data[feature].mean(), color = "green", linestyle = "--"
    )                   # Add mean to the histogram
    ax_hist2.axvline(
        data[feature].median(), color = "black", linestyle = "-"
    )                   # Add median to the histogram
```

Code Snippet 10. Call labeled barplot function for CHAS categorical variable

```python
labeled_barplot(df, "CHAS", perc = True)
```

Code Snippet 11. Call labeled barplot function for RAD categorical variable

```python
labeled_barplot(df, "RAD", perc = True)
```

Code Snippet 12. Call histogram-boxplot function for CRIM numerical variable

```python
histogram_boxplot(df, "CRIM")
```

Code Snippet 13. Call histogram-boxplot function for ZN numerical variable

```python
histogram_boxplot(df, "ZN")
```

Code Snippet 14. Call histogram-boxplot function for INDUS numerical variable

```
histogram_boxplot(df, "INDUS")
```

Code Snippet 15. Call histogram-boxplot function for NOX numerical variable

```
histogram_boxplot(df, "NOX")
```

Code Snippet 16. Call histogram-boxplot function for RM numerical variable

```
histogram_boxplot(df, "RM")
```

Code Snippet 17. Call histogram-boxplot function for AGE numerical variable

```
histogram_boxplot(df, "AGE")
```

Code Snippet 18. Call histogram-boxplot function for DIS numerical variable

```
histogram_boxplot(df, "DIS")
```

Code Snippet 19. Call histogram-boxplot function for TAX numerical variable

```
histogram_boxplot(df, "TAX")
```

Code Snippet 20. Call histogram-boxplot function for PTRATIO numerical variable

```
histogram_boxplot(df, "PTRATIO")
```

Code Snippet 21. Call histogram-boxplot function for LSTAT numerical variable

```
histogram_boxplot(df, "LSTAT")
```

Code Snippet 22. Call histogram-boxplot function for MEDV target numerical variable

```
histogram_boxplot(df, "MEDV")
```

Code Snippet 23. Plot Heatmap for all categorical and numerical variables

```
# Plot heatmap to show correlations between all variables
plt.figure(figsize = (12, 8))
cmap = sns.diverging_palette(230, 20, as_cmap = True)
sns.heatmap(df.corr(), annot = True, fmt = '.2f', cmap = cmap)
plt.show()
```

Code Snippet 24. Plot AGE vs. DIS Scatterplot

```
# Plot Scatterplot to visualize the relationship between AGE and DIS
plt.figure(figsize = (6, 6))
sns.scatterplot(x = 'AGE', y = 'DIS', data = df)
plt.show()
```

Code Snippet 25. Plot RAD vs. TAX Scatterplot

```
# Plot Scatterplot to visulaize the relationship between RAD and TAX
plt.figure(figsize = (6, 6))
sns.scatterplot(x = 'RAD', y = 'TAX', data = df)
plt.show()
```

Code Snippet 26. Adjusted correlation for RAD vs. TAX numerical variables

```
# Remove the data corresponding to high tax rate
df1 = df[df['TAX'] < 600]

# Import the required function
from scipy.stats import pearsonr

# Calculate the correlation
print("The correlation between TAX and RAD is " + str(round(pearsonr(df1['TAX'], df1['RAD'])[0], 2)) + ".")
```

Code Snippet 27. Plot INDUS vs. TAX Scatterplot

```
# Plot Scatterplot to visualize the relationship between INDUS and TAX
plt.figure(figsize = (6, 6))
sns.scatterplot(x = 'INDUS', y = 'TAX', data = df)
plt.show()
```

Code Snippet 28. Plot LSTAT vs. MEDV Scatterplot (target variable)

```
# Plot Scatterplot to visulaize the relationship between LSTAT and MEDV
plt.figure(figsize = (6, 6))
sns.scatterplot(x = 'LSTAT', y = 'MEDV', data = df)
plt.show()
```

Code Snippet 29. Plot RM vs. MEDV Scatterplot (target variable)

```
# Plot Scatterplot to visulaize the relationship between RM and MEDV
plt.figure(figsize = (6, 6))
sns.scatterplot(x = 'RM', y = 'MEDV', data = df)
plt.show()
```

Code Snippet 30. Plot INDUS vs. NOX Scatterplot

```
# Plot Scatterplot to visualize the relationship between INDUS and NOX
plt.figure(figsize = (6, 6))
sns.scatterplot(x = 'INDUS', y = 'NOX', data = df)
plt.show()
```

Code Snippet 31. Plot AGE vs. NOX Scatterplot

```
# Plot Scatterplot to visualize the relationship between AGE and NOX
plt.figure(figsize = (6, 6))
sns.scatterplot(x = 'AGE', y = 'NOX', data = df)
plt.show()
```

Code Snippet 32. Plot DIS vs. NOX Scatterplot

```
# Plot Scatterplot to visualize the relationship between DIS and NOX
plt.figure(figsize = (6, 6))
sns.scatterplot(x = 'DIS', y = 'NOX', data = df)
plt.show()
```

Code Snippet 33. Apply log-transformation to MEDV target variable

```
df['MEDV_log'] = np.log(df['MEDV'])
```

Code Snippet 34. Graph histogram of log-transformed MEDV target variable

```
sns.histplot(data = df, x = 'MEDV_log', kde = True)
```

Code Snippet 35. One-Hot encode RAD variable due to being categorical variable

```
# Convert RAD to a categorical variable
df['RAD_cat'] = df['RAD'].astype('category')
# One-hot encode the categories
RAD_dummies = pd.get_dummies(df['RAD_cat'], prefix='RAD', drop_first=True)
# Concatenate back to the dataframe
df = pd.concat([df, RAD_dummies], axis=1)
# Drop original columns
df = df.drop(columns=['RAD', 'RAD_cat'])
```

Code Snippet 36. Partition dataset into features (X) variables and target (Y) variable

```
# Separate the features (independent variables) and target column (dependent variable)
Y = df['MEDV_log']
X = df.drop(columns = {'MEDV', 'MEDV_log'})
# Add the intercept term
X = sm.add_constant(X)
```

Code Snippet 37. Splitting the dataset into a ratio of 80:20 train and test dataset

```
# Splitting the data in 70:30 ratio of train to test data
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size = 0.20, random_state = 1)
```

Code Snippet 38. Shape of the X train dataset

```
# Compute X_train rows and columns
X_train.shape
```

Code Snippet 39. Shape of the X test dataset

```
# Compute X_test rows and columns
X_test.shape
```

Code Snippet 40. Check for Multicollinearity Function

```
# Function to check VIF
def checking_vif(train):
    vif = pd.DataFrame()
    vif["feature"] = train.columns

    # Calculating VIF for each feature
    vif["VIF"] = [
        variance_inflation_factor(train.values, i) for i in range(len(train.columns))
    ]
    return vif

# Convert boolean columns to integer (0 or 1) before calculating VIF
# Identify boolean columns
bool_cols = X_train.select_dtypes(include='bool').columns
X_train_numeric = X_train.copy()
for col in bool_cols:
    X_train_numeric[col] = X_train_numeric[col].astype(int)


print(checking_vif(X_train_numeric))
```

Code Snippet 41. Print Multicollinearity VIF values (executed three times)

```
print(checking_vif(X_train_numeric))
```

Code Snippet 42. Drop RAD_24 Dummy Variable from Train dataset

```
# Drop RAD_24 variable and check for updated VIF score
X_train = X_train.drop('RAD_24', axis = 1)
```

Code Snippet 43. Drop TAX Dummy Variable from Train dataset

```
# Drop TAX variable and check for updated VIF score
X_train = X_train.drop('TAX', axis = 1)
```

Code Snippet 44. Drop RAD_24 and TAX Dummy Variables from Test dataset

```
# Drop RAD_24 and TAX variables in X_test
X_test = X_test.drop('RAD_24', axis = 1)
X_test = X_test.drop('TAX', axis = 1)
```

Code Snippet 45. Obtain updated Data Types for X train dataset

```
X_train.info()
```

Code Snippet 46. Obtain updated Data Types for X test dataset

```
X_test.info()
```

Code Snippet 47. Convert all Boolean Dummy Variables from the Training and Test datasets

```
# Convert boolean columns to integer (0 or 1) for X_train
bool_cols_train = X_train.select_dtypes(include='bool').columns
for col in bool_cols_train:
    X_train[col] = X_train[col].astype(int)

# Convert boolean columns to integer (0 or 1) for X_test
bool_cols_test = X_test.select_dtypes(include='bool').columns
for col in bool_cols_test:
    X_test[col] = X_test[col].astype(int)
```

Code Snippet 48. Compute First model (model1) summary

```
# Create the model
model1 = sm.OLS(y_train, X_train).fit()
# Get the model summary
model1.summary()
```

Code Snippet 49. Check First model (model1) Performance on X train and X test datasets

```
# Checking model performance
model_pref(model1, X_train, X_test)
```

Code Snippet 50. Drop all independent variables (features or predictors) from X train and X test datasets

```
# Drop all independent variables or features with p-values greater than 0.05 on X train and X test
X_train = X_train.drop(['INDUS', 'AGE', 'ZN', 'RAD_2', 'RAD_3', 'RAD_4', 'RAD_5', 'RAD_6', 'RAD_7', 'RAD_8'], axis = 1)
X_test = X_test.drop(['INDUS', 'AGE', 'ZN', 'RAD_2', 'RAD_3', 'RAD_4', 'RAD_5', 'RAD_6', 'RAD_7', 'RAD_8'], axis = 1)
```

Code Snippet 51. Compute Second model (model2) summary

```
# Create the model
model2 = sm.OLS(y_train, X_train).fit()
# Get the model summary
model2.summary()
```

Code Snippet 52. Check Second model (model2) Performance on X train and X test datasets

```
# Checking model2 performance
model_pref(model2, X_train, X_test)
```

Code Snippet 53. Check mean of residuals for Second model (model2)

```
residuals = model2.resid
np.mean(residuals)
```

Code Snippet 54. Check for homoscedasticity of residuals for Second model (model2)

```
# Check for homoscedasticity using Goldfeld Quandt test model2
name = ["F statistic", "p-value"]
test = sms.het_goldfeldquandt(y_train, X_train)
lzip(name, test)
```

Code Snippet 55. Graph Residual Plot to check for linearity of variables for Second model (model2)

```
# Plot residuals model2
fitted = model2.fittedvalues
sns.residplot(x = fitted, y = residuals, color = "lightblue", lowess = True)
plt.xlabel("Fitted Values")
plt.ylabel("Residual")
plt.title("Residual PLOT")
plt.show()
```

Code Snippet 56. Graph Residuals Q-Q plot to check for normality of residuals for Second model (model2)

```
# Plot q-q plot of residuals model2
stats.probplot(residuals, dist = "norm", plot = pylab)
plt.show()
```

Code Snippet 57. Plot Residuals Histogram to check for normality of residuals for Second model (model2)

```
# Plot histogram of residuals model2
sns.histplot(residuals, kde = True)
```

Code Snippet 58. Apply Cross-validation to improve Second model (model2)

```
# Build the regression model and cross-validate
linearregression = LinearRegression()

cv_Score11 = cross_val_score(linearregression, X_train, y_train, cv = 10)
cv_Score12 = cross_val_score(linearregression, X_train, y_train, cv = 10,
                             scoring = 'neg_mean_squared_error')


print("RSquared: %0.3f (+/- %0.3f)" % (cv_Score11.mean(), cv_Score11.std() * 2))
print("Mean Squared Error: %0.3f (+/- %0.3f)" % (-1*cv_Score12.mean(), cv_Score12.std() * 2))
```

Code Snippet 59. Obtain coefficients for independent variables (predictors) for Second Model (model2)

```
# Obtain the coefficients for model2
coef = model2.params
pd.DataFrame({'Feature' : coef.index, 'Coefs' : coef.values})
```